

Niayesh Afshordi

Astrophysics, cosmology, and gravity

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Comprehensive public academic CV — updated July 2026

Profile

Niayesh Afshordi is an astrophysicist working at the interface of cosmology, gravitation, quantum theory, and observation. His research program connects deep questions in fundamental physics to gravitational-wave data, the cosmic microwave background, large-scale structure, compact objects, and other astrophysical probes. Current themes include quantum aspects of black holes, gravitational-wave echoes and strong-gravity tests, early-universe cosmology, dark matter and dark energy, missing baryons, and the observational consequences of modified gravity.

Academic appointments

Full Professor, Department of Physics and Astronomy, University of Waterloo	2022–present
Professor in the Astrophysics and Gravitation group; founding member of the Waterloo Centre for Astrophysics (2018).	
Research Associate Faculty, Perimeter Institute for Theoretical Physics	2009–present
Associate Faculty in the Cosmology and Gravitation group.	
Associate Professor, University of Waterloo	2015–2022
Assistant Professor, University of Waterloo	2009–2015
Distinguished Research Fellow, Perimeter Institute	2007–2009
Postdoctoral Fellow, Institute for Theory and Computation, Harvard–Smithsonian Center for Astrophysics	2004–2007

Education

PhD, Astrophysics, Princeton University	2000–2004
Thesis: <i>The Other 99 Percent</i> .	
BSc, Physics, Sharif University of Technology	1996–1999

Selected distinctions

Buchalter Cosmology Prize	2019; 2015
Professor M. K. Vainu Bappu Gold Medal, Astronomical Society of India	2011
Ontario Early Researcher Award	2011
NSERC Discovery Accelerator Supplement	2010–2013

Selected research support

NSERC Discovery Grant: “Quantum Black Holes, Vacuum, and the Big Bang: Unraveling cosmic mysteries with Multi-Messenger Astronomy”	2025–2030
Perimeter Travel and Technology Fund	2009–2030
University of Waterloo Interdisciplinary Trailblazer Grant: “Quantum Spacetime in the Cosmos: From Conception to Reality”	2022–2025
NSERC Discovery Grant: “Novel Astrophysical Probes of Fundamental Physics”	2020–2025

Scope and sources

This public version is designed for reading as well as completeness. The professional record is generated from the Canadian Common CV submitted on 15 August 2025, with protected personal fields excluded. The final annex inventories the locally maintained talks archive. Publications, presentations, students, postdoctoral fellows, funding, service, outreach, and other professional activities are retained in the sections below. Current group roles and destinations are reviewed before each scheduled update.

Complete professional record

Generated from the Canadian Common CV submitted on 15 August 2025. The protected cover page and personal contact, demographic, citizenship, and identifier fields are excluded. Administrative form metadata is suppressed for readability; the substantive academic record is retained.

Privacy screen: 3 protected source line(s) removed in addition to the protected cover page.

Language Skills

English and Persian: read, write, speak, understand, and peer review.

Degrees

2000/9 - 2004/11	Doctorate, PhD, Astrophysics, Princeton University Thesis Title: The Other 99 Percent Supervisors: Spergel, David, 2000/9 - 2004/11 Research Disciplines: Astronomy and Astrophysics
1996/9 - 1999/5	Bachelor’s Honours, BSc, Physics, Sharif University of Technology

Recognitions

2020/1	The Buchalter Cosmology Prize 2019 Dr. Ari Buchalter Prize / Award The \$10,000 First Prize was awarded to Dr. Jahed Abedi of the Max-Planck-Institut fur Gravitationsphysik (Albert-Einstein-Institut) and Leibniz Universitat Hannover and Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics, for their work entitled "Echoes from the Abyss: A highly spinning black hole remnant for the binary neutron star merger GW170817", published in the Journal of Cosmology and Astroparticle Physics and recognized by the judging panel as “a bold and innovative step towards understanding quantum gravitational phenomena, using observational data to demonstrate gravitational wave echoes from a binary neutron star merger, which could indicate that black hole interiors may be more complex than the simple regions predicted by General Relativity.”
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2016/1	The Buchalter Cosmology Prize 2015 Dr. Ari Buchalter Prize / Award The \$2,500 Third Prize was awarded to Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics and Dr. Elliot Nelson of Perimeter Institute, for their work entitled "Cosmological Non-Constant Problem: Cosmological bounds on TeV-scale physics and beyond", published in the Journal of Physical Review D and recognized by the judging panel as "an intriguing proposition that the Planck scale of quantum gravity may soon be accessible by particle accelerators, based on the expected white noise fluctuations in the cosmological constant, which might not be constant at all but vary widely in space."
2011/10	Professor M.K.Vainu Bappu Gold Medal Astronomical Society of India Prize / Award The Vainu Bappu Memorial Gold Medal honours outstanding astronomers worldwide under the age of 35. I was one of the two recipients of this honour in 2011.
2011/9	Early Researcher Award - 150,000 (Canadian dollar) Government of Ontario Prize / Award The ERA program helps promising, recently-appointed Ontario researchers build their research teams of undergraduates, graduate students, post-doctoral fellows, research assistants, associates and technicians. The goal of the program is to improve Ontario's ability to attract and retain the best and brightest research talent.
2010/9 - 2013/9	Discovery Accelerator Supplement - 120,000 Natural Sciences and Engineering Research Council of Canada (NSERC) Prize / Award The Discovery Accelerator Supplement award provides substantial and timely resources to a small group of researchers whose research proposals suggest and explore highrisk, novel or potentially transformative concepts and lines of inquiry, and are likely to have impact by contributing to groundbreaking advances in the area. I was one of the 8 researchers in Physics who received this award in 2010.

User Profile

Researcher Status: Researcher Research Career Start Date: 1998/01/01 Engaged in Clinical Research?: No

Key Theory / Methodology General Relativity, Quantum Mechanics, Astrophysical and Cosmological Observations, Numerical Simulations, Large Data, Statistical Methods, Theoretical Modeling

Research Interests I lead a comprehensive and ambitious research program, which spans a wide area, from Astrophysics/Cosmology to Fundamental Physics, and includes a diverse spectrum of observational, numerical, and analytic elements. The program aims at linking deep mysteries in fundamental physics to novel observational results and techniques, which can be used to shed light on the structure and laws of our Universe. The areas that my group and I currently focus on are: 1. Phenomenology of Dark Energy and cosmological modifications of General Relativity 2. Distribution of Dark Matter on small scales and novel windows for Dark Matter detection 3. Observational consequences of Aether theories of gravity, and their connection to Quantum Gravity and Cosmological Constant Problem 4. Early Universe, and its links to foundations of Gravity and Quantum Mechanics 5. Missing baryons, and observational probes of intergalactic plasma 6. Physics of Neutron Stars, Black Holes and Accretion Disks.

Research Experience Summary I am an Astronomer, and a Physicist; in short I am an Astrophysicist. Currently, I am a Professor in the Astrophysics and Gravitation group at the Department of Physics and Astronomy, University of Waterloo. I am also an Research Associate Faculty in the Cosmology and Gravitation group at the Perimeter Institute for Theoretical Physic (PI), the premier Canadian institute for research in theoretical Physics. I finished my PhD at Princeton University Observatory in August 2004. I was then a fellow at the Institute for Theory and Computation (ITC) at the Harvard-Smithsonian Center for Astrophysics (2004-2007), and a Distinguished Research Fellow at PI (2008-2009). To date, I have published more than 120 peer-reviewed articles, which have been cited more than 7600 times. I have also (co-)supervised 86 students, and 28 postdoctoral fellows. Moreover, I have co-organized 31 international scientific meetings/seminar series, and raised over 1.5 million dollars in external funding.

Research Specialization Keywords Astrophysics, Cosmology, Gravitational Waves, Physics of Gravity, Cosmic Microwave Background, Dark Matter, Dark Energy, Black Hole Physics and Astrophysics, Missing Baryons, Lorentz Violation, Quantum Gravity, COVID-19 epidemeology, Flash Radiotherapy

Disciplines Trained In Astronomy and Astrophysics, Physics

Research Disciplines Astronomy and Astrophysics, Physics, Epidemiology and Biostatistics, Radiology

Areas of Research Galactic and Cosmic, Universe Structure, Gravitation, Quantum Phenomena, Elementary

Particles

Fields of Application Foundations and Knowledge Acquisition, Biomedical Aspects of Human Health, Public Health

Employment

2022/9	Full Professor of Physics and Astronomy Physics and Astronomy, University of Waterloo Full-time, Professor Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)
2009/9	Research Associate Faculty Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics A Renewable appointment as a Perimeter Institute for Theoretical Physics Faculty Member.
2015/9 - 2022/7	Associate Professor Physics and Astronomy, Waterloo Centre for Astrophysics, University of Waterloo Full-time, Associate Professor Associate Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)
2009/9 - 2015/9	Assistant Professor Physics and Astronomy, Science, University of Waterloo Full-time, Assistant Professor Assistant Professor in the Astrophysics and Gravitation group
2012/9 - 2013/8	Visiting Professor Physics, North, SUNY at Buffalo Part-time, Visiting Professorship I visited this department on a regular basis and engaged and collaborated with the faculty in Cosmology.
2007/9 - 2009/8	Distinguished Research Fellow Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics
2004/9 - 2007/8	Postdoctoral Fellow Astrophysics, Institute for Theory and Computation, Harvard-Smithsonian Center for Astrophysics
2000/9 - 2004/8	Graduate Student Instructor and Researcher Astrophysical Sciences, Arts and Sciences, Princeton University
1999/9 - 2000/8	Graduate Student Teaching Assistant Physics, Arts and Sciences, Brown University
1998/9 - 1999/8	Research Associate Physics, Tehran, Institute for Theoretical Studies in Physics and Mathematics (IPM)
1996/1 - 1998/8	Instructor Iranian National Physics Olympiad Camp, International Physics Olympiad

Affiliations

The primary affiliation is denoted by (*)

(*) 2015/9	Associate Professor, University of Waterloo
2009/9	Associate Faculty, Perimeter Institute for Theoretical Physics

Leaves of Absence and Impact on Research

2019/9 - 2020/8	Sabbatical, University of Waterloo I was on sabbatical at Perimeter Institute, and did not have any research/teaching duties at U-Waterloo.
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Research Funding History

Awarded

2009/9 - 2030/7	Perimeter Travel and Technology Fund (ongoing), Grant Principal Investigator Funding Sources: Perimeter Institute Travel and Technology Fund
2025/4 - 2030/3	Quantum Black Holes, Vacuum, and the Big Bang: Unraveling cosmic mysteries with Principal Investigator Multi-Messenger Astronomy, Grant
2025/4 - 2030/3	Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant
2022/9 - 2025/8	Quantum Spacetime in the Cosmos: From Conception to Reality, Grant Principal Investigator Funding Sources: University of Waterloo Interdisciplinary Trailblazer grant Co-investigator : Fraser, Doreen; Collaborator : Weinstein, Steven
2020/4 - 2025/3	Novel Astrophysical Probes of Fundamental Physics, Grant Principal Applicant Funding Sources: Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery

Completed

2021/9 - 2023/8	HQP Pooled Resources Competition, Grant Principal Investigator Funding Sources: Arthur B. McDonald Canadian Astroparticle Physics Research Institute HQP Pooled Resources Competition
2015/4 - 2020/3	Novel Astrophysical Probes of Fundamental Physics, Grant Principal Investigator Funding Sources: Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery
2011/4 - 2016/3	Early Researcher Award, Grant Principal Applicant
2011/9 - 2016/9	Ontario Ministry of Research and Innovation (ON) Early Researcher Award
2010/4 - 2015/3	Discovery Grant, Grant Principal Applicant Funding Sources:
2010/4 - 2015/3	Natural Sciences and Engineering Research Council of Canada (NSERC)
2010/4 - 2013/3	Discovery Accelerator Supplement, Grant Principal Applicant Funding Sources:
2010/4 - 2013/3	Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Accelerator Supplement

Courses Taught

Instructor, Science, University of Waterloo Course Title: Introductory Astronomy Course Code: 238 Course Level: Undergraduate Academic Session: Winter Number of Students: 130 Lecture Hours Per Week: 3

Student/Postdoctoral Supervision

Bachelor's

2023/9 - 2025/8	Rasheed, Nabeel (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Analytic models for gravitational wave form templates" Present Position: PhD Student, Massachusetts Institute of Technology
2023/9 - 2024/7	Yin, Xitai (John) (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Black Holes in Quantum Gravity Present Position: Master's student in Physics, University of Waterloo
2023/9 - 2025/8	Tang, Ruijing (In Progress) , University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Exploring Gravitational Effects on accretion disk dynamics in the Teleparallel Born-Infeld gravity" Present Position: BSc Student, University of Waterloo

2023/8 - 2021/9	Liu, Ruolin (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Detecting cosmological scalar fields using orbital networks of quantum sensors" Present Position: MSc Student, University of Waterloo
2022/1 - 2025/4	Subhi, Hassan (In Progress) , University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/4 Thesis/Project Title: Research Project: "Gravitational Wave Echoes in Pulsar Timing Arrays" Present Position: Undergraduate Student, University of Waterloo
2021/2 - 2021/9	Zheng, Jolene (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Covid forecast modeling and online dashboard Present Position: Software Engineer, Bloomberg
2021/2 - 2021/9	Olalekan Koiki, Shafika (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Co-op project: "Covid forecast modeling and online dashboard" Present Position: none
2021/1 - 2021/8	Wisniewski, Kasia (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Searching for Black Hole Echoes from the Failed Supernova AT2017be" Present Position: Operations Technician, GHGSat Inc

Bachelor's Honours

2024/9 - 2025/5	Shah, Saral (In Progress) , University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Dynamical Friction in a neutrinos sea" Present Position: Undergraduate Student, University of Waterloo
2024/9 - 2025/5	Yanover, Lyvia (In Progress) , University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Bi-Thermal Big Bang" Present Position: Undergraduate Student, University of Waterloo
2024/9 - 2025/5	Murali, Nikhil (In Progress) , University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's Thesis: "Quantum Black Holes: quantum corrections at the tip of a cigar" Present Position: Undergraduate Student, University of Waterloo
2024/9 - 2025/5	Brisebois, Alex Dominic Raul (In Progress) , University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Clustering in the Action-Angle Space" Present Position: Undergraduate Student, University of Waterloo
2023/9 - 2024/1	Zhao, Guandi (Completed) , Fudan University, China Principal Supervisor Thesis/Project Title: BSc visiting student at Perimeter Institute for Theoretical Physics: "Thermal SZ contribution from Low mass galaxies in the unWISE survey" Present Position: MSc student, ETH Zurich, Switzerland
2023/9 - 2024/8	Goel, Lakshay (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Accelerated detector in a superposed spacetime Present Position: none
2023/9 - 2024/7	Gifford, Hazel (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Bohmian Rhapsody: Cuscuton and Cosmology Present Position: MSc Student, Brock University
2022/9 - 2023/6	Schiebelbein, Aryanna (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Projects: "Mystery noise in LIGO" Present Position: PhD Student, University of Toronto
2022/1 - 2024/7	Li, Yu (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Detecting cosmological scalar fields using orbital networks of quantum sensors" Present Position: PhD Student, California Institute of Technology
2021/5 - 2023/8	Lawrence, Jensen Pierce Thomas (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Research Project: "Universal horizon formation in the incompressible limit of Einstein-Aether theory" Present Position: Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, PhD Student
2020/9 - 2023/8	Wen, Robin (Completed) , Department of Applied Math, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis project: "A Cosmic Glitch in gravity" Present Position: PhD Student, California Institute of Technology

2020/9 - 2021/8	Stewart, Meagan (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Inferring Transient Weak Lensing from Quasar Light Curves" Present Position: MSc Student, Simon Fraser University, British Columbia
2020/5 - 2022/5	Shawqi, Shafayat (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Sorkin-Johnston Effective Field Theory" Present Position: PhD Student, University of Alberta
2020/1 - 2020/5	Duff, Petra Mary (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Echoes from the Abyss" Present Position: PhD Student, University of Toronto
2019/9 - 2022/8	Minhas, Palka (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Fast Radio Bursts Present Position: MSc Student, McGill University
2017/1 - 2017/4	Adair, Connor (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Gravitational Wave Echoes Present Position: MSc Student, University of British Columbia
2016/1 - 2016/4	Dykaar, Hannah (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA research project: "Echoes from the Abyss" Present Position: PhD Student, University of Toronto
2016/1 - 2016/4	Elgood, Zachary (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Einstein-Aether Black Holes Present Position: La Caixa Doctoral Scholar, Institute of Theoretical Physics (IFT), Madrid, Spain
2010/9 - 2011/6	Bovard, Luke (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Undergraduate B.Sc. Honours Research Project Present Position: Postdoctoral Researcher, Goethe University, Germany
2008/9 - 2009/8	Mueller, Matthias (Completed) , Ulm University Co-Supervisor Thesis/Project Title: Undergraduate Research Project Present Position: Postdoctoral Fellow, University of Florence

Master's non-Thesis

2023/9 - 2025/8	Liu, Ruolin (In Progress) , University of Waterloo Principal Supervisor Thesis/Project Title: Quadratic Gravity in Big Bang and Black Holes Present Position: MSc Student, University of Waterloo
2023/3 - 2023/12	Friedman-Shaw, Batia (Completed) , Perimeter Institute for Theoretical Physics/ University Principal Supervisor of Waterloo Thesis/Project Title: MSc Project:"Doppler Bias: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys" Present Position: PhD Student, Perimeter Institute for Theoretical Physics/ University of Waterloo
2022/9 - 2023/6	Hrytseniak, Oleksandra (Completed) , Perimeter Institute for Theoretical Physics/ Co-Supervisor University of Waterloo Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity" Present Position: PhD student, Perimeter Institute for Theoretical Physics/ University of Waterloo
2019/9 - 2020/8	Chua, Wan Zen (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: PSI Research Project: "Electromagnetic Albedo of Fuzzballs" Present Position: PhD student, Cornell University
2015/9 - 2016/6	Wang, Qingwen (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Fuzzball approach to black holes Present Position: PhD Student, University of Waterloo
2013/8 - 2014/6	Roberts, Brodyn (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: PSI Research Project Present Position: Software Engineer, Index Exchange, Toronto, Canada
2010/9 - 2012/6	Agarwal, Abhineet (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Thermal Tachyacoustic Cosmology Present Position: PhD Student, Jamia Millia University, India

Master's Thesis

2022/9 - 2024/8	Yang, Jiayue (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Entropy of Analogue Black Holes in Tilted Dirac Cone Materials Present Position: MSc Student, University of Waterloo
2021/1 - 2021/12	Golafshani, Hamid (Withdrawn) , University of Waterloo Principal Supervisor Thesis/Project Title: Topics in Astrophysics and Cosmology Present Position: Professor of Mechanical Engineering, Conestoga College, Waterloo, ON
2018/9 - 2020/8	Hergott, Samantha (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Unifying Big Bang and Black Holes Present Position: PhD Student, York University
2018/9 - 2019/8	Yang, Tianyi (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Cosmic Filaments, Galactic Kinematics in Gaia Present Position: PhD Student, University of Edinburgh
2018/9 - 2020/8	Chen, Alice (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Cosmological Structure Formation Present Position: PhD Student, University of Waterloo
2017/9 - 2018/7	Boruah, Supranta (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Constraining the mass of Milky Way based on clustering in the action space of Gaia DR2 stars Present Position: Postdoctoral Fellow, University of Pennsylvania
2016/9 - 2017/8	Kim, Hyungjin (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: The Early Universe: New aspects of Cuscuton Theory, Modified Dispersion Relations and Pulsar Constraints on Vacuum Noise Present Position: PhD Student, University of Waterloo
2014/9 - 2016/8	Meiers, Michael (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Universal Horizons in Black Holes Present Position: PhD Student, University of California-Davis
2012/8 - 2013/6	Okoli, Chiamaka (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Concentration of cosmological haloes Present Position: Deceased
2011/9 - 2013/5	Yazdi, Yasaman (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: A Spacetime Approach to Defining Vacuum States and Entropy Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta
2011/8 - 2012/6	Schuhmann, Robert (Completed) , University of Waterloo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: PSI Research Project Present Position: Postdoctoral Research Associate, University of Manchester
2009/8 - 2010/6	Ponce, Yosdanis (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Cosmology of a Black Hole Lattice Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus

Doctorate

2023/10 - 2025/8	Nasiri, Arad (In Progress) , Imperial College London Co-Supervisor Student Degree Expected Date: 2025/8 Thesis/Project Title: Project Supervision: "Stochastic Dark Matter: Covariant Brownian Motion from Planckian Discreteness (arXiv:2409.02188)" Present Position: PhD Student, Imperial College London
2023/9 - 2024/12	Patterson, Everett Avison (In Progress) , University of Waterloo Co-Supervisor Student Degree Expected Date: 2025/12 Thesis/Project Title: Project Supervision: "Accelerated detector in a superposed spacetime (arXiv:2409.12441)" Present Position: PhD Student, University of Waterloo
2023/1 - 2026/12	Chen, Alice (In Progress) , University of Waterloo and Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Machine Learning and Cosmic Structure Formation Present Position: PhD Student

2022/9 - 2025/8	Borissova, Johanna (In Progress) , Perimeter Institute for Theoretical Physics/ University Co-Supervisor of Waterloo Thesis/Project Title: Scale-invariance at the core of quantum black holes Present Position: PhD Student, Perimeter Institute for Theoretical Physics/ University of Waterloo
2022/7 - 2024/12	Kim, Leo (In Progress) , Queen's University Co-Supervisor Thesis/Project Title: Project Supervision: "A Poisson Log-Normal Framework for Dark Sirens" Present Position: PhD Student, Queen's University
2022/7 - 2022/9	Boyanov Savov, Valentin (Completed) , Universidad Complutense de Madrid Principal Supervisor Thesis/Project Title: Visiting PhD student: "Aspects of Quantum Black Holes in semiclassical gravity" Present Position: Postdoctoral Fellow, Instituto Superior Tecnico, Lisbon, Portugal
2020/9 - 2024/12	Dailey, Conner (All But Degree) , University of Waterloo and Perimeter Institute Principal Supervisor Student Degree Expected Date: 2024/12 Thesis/Project Title: On the Initial Boundary Value Problem in Numerical Relativity Present Position: PhD student, University of Waterloo and Perimeter Institute
2019/8 - 2020/4	Longo, Luis Felipe (Completed) , University of Sao Paolo, Brazil and Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: visiting PhD student: "Echo-Diversity in Binary Black Hole Mergers" Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich- Schiller Universitaet, Jena, Germany
2019/2 - 2020/7	Tsuna, Daichi (Completed) , University of Tokyo Co-Supervisor Thesis/Project Title: visiting PhD student: "Quantum Black Hole Seismology" Present Position: Sherman Fairchild Postdoctoral Scholar Research Associate, California Institute of Technology
2018/9 - 2019/8	Held, Aaron (Completed) , University of Heidelberg Principal Supervisor Thesis/Project Title: Visiting PhD student: "Quantum Black Holes in Asymptotic Safety" Present Position: Postdoctoral Fellow, Ecole Normale Supérieure, Paris, France
2018/9 - 2019/8	Giri, Utkarsh (Completed) , University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Pressure of haloes in cosmological simulations Present Position: Postdoctoral Fellow, California Institute of Technology
2018/9 - 2023/8	Saraswat, Krishan (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Quantum Aspects of Black Holes: From Microstates to Echoes and Somewhere In-Between Present Position: Postdoctoral Fellow, University of Southern California
2018/1 - 2018/9	Oshita, Naritaka (Completed) , University of Tokyo Principal Supervisor Thesis/Project Title: visiting PhD student: "Probing microstructure of black hole spacetimes with gravitational wave echoes" Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto, Japan
2017/9 - 2018/4	Dhumuntarao, Aditya (Completed) , University of Minnesota Principal Supervisor Thesis/Project Title: visiting NSF PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime" Present Position: Postdoctoral Researcher, Sandia National Laboratories
2017/9 - 2021/6	Robbins, Matthew (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Gravitational Waves, Bose-Einstein Condensates, and Modified Gravity Present Position: Teaching Faculty (tenure-track), University of Toronto
2017/9 - 2018/4	Gherzi, Jose Tomas Galvez (Completed) , Simon Fraser University Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime" Present Position: Assistant Professor, Universidad de Ingenieria y Tecnologia, Peru
2016/9 - 2020/6	Wang, Qingwen (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Black Hole Echology Present Position: Senior Quant, Deloitte Business Consulting and Services

2016/9 - 2018/8	Abedi, Jahed (Completed) , Sharif University of Technology Co-Supervisor Thesis/Project Title: Black Hole Echoes Present Position: Postdoctoral Fellow, University of Stavanger, Norway
2016/9 - 2017/8	Haghighi, Mohammad Zhooldideh (Completed) , Sharif University Principal Supervisor Thesis/Project Title: visiting PhD student: "X-ray Cooling Flows and Modified Gravity" Present Position: Postdoctoral Fellow, IPM, Tehran
2015/9 - 2016/8	Lynch, Morgan (Completed) , University of Wisconsin-Milwaukee Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperatures of renormalizable quantum field theories in curved spacetime" Present Position: Zuckerman Postdoctoral Scholar, Technion
2014/9 - 2018/6	Karami, Mansour (Completed) , University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Dark Matter on Small Scales Present Position: Senior Machine Learning Engineer, Riskfuel, Toronto
2014/9 - 2018/6	Altamirano, Natacha (Completed) , University of Waterloo/Perimeter Institute for Co-Supervisor Theoretical Physics Thesis/Project Title: The quantum and the Gravity: Newtonian and Cosmological Applications Present Position: Director, Data Science, Birch Hill Equity Partners, Toronto, Canada
2014/9 - 2018/6	Okoli, Chiamaka (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Dark Matter and Neutrinos in the Foggy Universe Present Position: Deceased
2014/4 - 2014/11	Nelson, Elliott (Completed) , Penn State University Principal Supervisor Thesis/Project Title: Cosmological Non-Constant Problem Present Position: MSc Student in Computer Science, University of Waterloo, Canada
2013/9 - 2017/6	Yazdi, Yasaman (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Physics and Astrophysics of Black Holes Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta
2013/9 - 2014/8	Narimani, Ali (Completed) , University of British Columbia Principal Supervisor Thesis/Project Title: visiting PhD student: "How does pressure gravitate?" Present Position: Data Scientist, Twitter
2013/9 - 2014/6	Karami, Mansour (Completed) , Sharif University of Technology Co-Supervisor Thesis/Project Title: Cosmic Structure Formation, Modified Gravity and Dark Energy, Gravitational Lensing Present Position: Manager, Market Risk Measurement, Scotiabank, Canada
2013/9 - 2017/6	Zwane, Nosiphiwo (Completed) , University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Cosmological Tests of Causal Set Phenomenology Present Position: Lecturer, University of Eswatini
2013/9 - 2014/8	Jeremy Sakstein (Completed) , University of Cambridge Co-Supervisor Student Degree Start Date: 2010/9 Student Degree Received Date: 2014/7 Thesis/Project Title: visiting PhD student: "Novel Approaches to Modified Gravity" Present Position: Postdoctoral Fellow, Center for Particle Cosmology, University of Pennsylvania, USA
2011/9 - 2017/6	Gould, Elizabeth (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics? Present Position: Postdoctoral Fellow, Arthur B. McDonald Canadian Astroparticle Physics Research Institute
2011/9 - 2016/6	Saravani, Mehdi (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Aspects of Non-locality Present Position: Corporate Treasury Strategist, Goldman-Sachs, London, UK
2011/9 - 2014/7	Pourhasan, Razieh (Completed) , University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Adventures in Holography Present Position: Research Assistant, University of Ottawa, Canada

2011/9 - 2015/5	Kamiab, Farbod (Completed) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Neutron Stars, the Exotica - From Modifying General Relativity to Strong Magnetic Fields Present Position: Analytics, Facebook, Dublin, Ireland
2011/1 - 2012/8	Ponce, Yosdanis (Withdrawn) , University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Student Degree Start Date: 2011/1 Thesis/Project Title: Black Hole Halo Model/ Holographic Cosmology Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus
2010/9 - 2011/9	Baghran, Shant (Completed) , Sharif University of Technology Co-Supervisor Thesis/Project Title: The Observational Tests of Dark Energy and Modified Gravity Theories Present Position: Assistant Professor, Sharif University, Tehran, Iran
2010/9 - 2011/6	Gryb, Sean (Completed) , University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Shape Dynamics and Mach's Principles: Gravity from Conformal Geometrodynamics Present Position: Research Associate, Philosophy Department, University of Bristol, England
2010/9 - 2014/6	Aslanbeigi, Siavash (Completed) , University of Waterloo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: Cosmic Atoms: from Causal Sets to Clusters Present Position: Vice President, Trading Products, BMO Capital Markets, Canada
2008/9 - 2010/6	Prescod-Weinstein, Chanda (Completed) , University of Waterloo Co-Supervisor Thesis/Project Title: Cosmic Acceleration As Quantum Gravity Phenomenology Present Position: Assistant Professor, University of New Hampshire
2007/9 - 2008/8	LoVerde, Marilena (Completed) , Columbia University Co-Supervisor Thesis/Project Title: Extended Limber Approximation Present Position: Assistant Professor, Stony Brook University, New York

Post-doctorate

2024/3 - 2024/12	Dunsky, David (In Progress) , New York Univesrity Co-Supervisor Thesis/Project Title: Project Supervision: "Dark Matter Disk Accretion into Black Holes" Present Position: Postdoctoral Fellow, New York Univesrity
2024/3 - 2024/12	Ghalsasi, Akshay (In Progress) , Harvard University Co-Supervisor Thesis/Project Title: Project Supervsion: "Dark Matter Disk Accretion onto Black Holes" Present Position: Postdoctoral Fellow, Deoartment of Physics, Harvard University
2024/3 - 2026/3	Faraji, Shokoufe (In Progress) , University of Waterloo Principal Supervisor Thesis/Project Title: Quantum Gravity and Accretion Physics around Black Holes Present Position: Postdoctoral Fellow, Department of Physics and Astronomy, University of Waterloo
2023/3 - 2024/12	Krolewski, Alex (In Progress) , University of Waterloo/ Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys Present Position: CITA National Fellow, Waterloo Centre for Astrophysics
2023/1 - 2025/9	Quintin, Jerome (In Progress) , University of Waterloo Co-Supervisor Thesis/Project Title: Quadratic Gravity in Cosmology and Black Holes Present Position: Faculty of Mathematics William T. Tutte Postdoctoral Fellow, Department of Applied Mathematics, University of Waterloo
2022/8 - 2024/12	Hassani, Farbod (In Progress) , University of Oslo Principal Supervisor Thesis/Project Title: Project Supervision: "Simulating dynamical friction with cosmic neutrinos" Present Position: Postdoctoral Fellow, University of Oslo
2022/1 - 2024/12	Longo, Luis Felipe (In Progress) , Friedrich-Schiller Universitaet, Jena, Germany Principal Supervisor Thesis/Project Title: Spectrogram Correlated Stacking Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich- Schiller Universitaet, Jena, Germany

2021/9 - 2022/8	Muir, Jessica (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity" Present Position: Postdoctoral Fellow, Perimeter Institute for Theoretical Physics
2021/3 - 2025/4	Mylova, Maria (In Progress) , Perimeter Institute for Theoretical Physics and IPMU, Japan Principal Supervisor Thesis/Project Title: 1- Non-Gaussian Signatures of a Thermal Big Bang 2- Effective Cuscuton Theory Present Position: Emmy Noether visiting postdoctoral fellow, Perimeter Institute for Theoretical Physics
2020/10 - 2024/12	Nikakhtar, Farnik (In Progress) , Yale University Co-Supervisor Thesis/Project Title: Project Supervision: "Action-Angle Space Clustering" Present Position: Postdoctoral Fellow, Yale University
2020/3 - 2024/3	Dey, Ramit (Completed) , Western University, Ontario, Canada Co-Supervisor Thesis/Project Title: 1- Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background 2- Black Hole Echoes Present Position: Research Associate, Western University, Ontario, Canada
2020/2 - 2020/6	Shoshani, Barak (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Bayesian Search for Black Hole Echoes Present Position: Assistant Professor, Physics Department, Brock University
2019/4 - 2021/3	Oshita, Naritaka (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Echoes from Quantum Black Holes Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto
2019/1 - 2023/12	Abedi, Jahed (Completed) , University of Stavanger, Norway Principal Supervisor Thesis/Project Title: Search for Black Hole Echoes in Gravitational Wave Observations Present Position: Assistant Professor, Beijing Institute for Mathematical Sciences and Applications (BIMSA)
2015/9 - 2018/6	Nelson, Elliot (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Quantum Decoherence during Inflation from Gravitational Nonlinearities Present Position: Research Scientist, IBM Research
2013/9 - 2017/8	Guariento, Daniel (Completed) , Universidade de Sao Paulo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: Cosmological Black Holes from Self-Gravitating Fields Present Position: Professor, Conestoga College, Canada
2013/9 - 2014/8	Scrimgeour, Morag (Completed) , University of Waterloo Principal Supervisor Thesis/Project Title: Mass of Neutrinos from Neutrino Wakes Present Position: Data Scientist, Square, San Francisco CA
2012/9 - 2015/8	Marsh, David (Completed) , Perimeter Institute for Theoretical Physics Thesis/Project Title: Structure of Dark Matter Haloes Present Position: Sofja Kovalevskaja Fellow, University of Goettingen
2012/9 - 2016/8	Kubiznak, David (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Black Hole Chemistry Present Position: Senior Fellow, Perimeter Scholar International program
2012/9 - 2016/8	Hinterbichler, Kurt (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Consistent modifications of Einstein gravity Present Position: Assistant Professor, Case-Western Reserve University
2012/9 - 2015/8	Kobayashi, Takeshi (Completed) , Perimeter Institute for Theoretical Physics/Canadian Principal Supervisor Institute for Theoretical Astrophysics Thesis/Project Title: Primordial Magnetic Fields from the Post-Inflationary Universe Present Position: Postdoctoral Fellow, ICTP Italy
2011/9 - 2013/8	Lavaux, Guilhem (Completed) , University of Waterloo/Perimeter Institute-CITA National Co-Supervisor Fellow Thesis/Project Title: A New Light on Missing Baryons Present Position: CNRS Permanent Position, L'Institut d'Astrophysique de Paris, France

2011/9 - 2012/8	Baghrām, Shant (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Prospects for Detecting Dark Matter Halo Substructure with Pulsar Timing Present Position: Professor at Sharif University, Sharif University, Tehran, Iran
2010/9 - 2013/8	Zavala, Jesus (Completed) , University of Waterloo - CITA National Fellow Principal Supervisor Thesis/Project Title: Clustering in the Phase Space of Dark Matter Haloes Present Position: Professor at U-Iceland, University of Iceland
2010/9 - 2012/8	Johnson, Matthew (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Determining the Outcome of Cosmic Bubble Collisions in Full General Relativity Present Position: Associate Professor at York University, Associate Faculty at Perimeter Institute
2009/9 - 2011/8	Shandera, Sarah (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Breakdown of Semiclassical Methods in de Sitter Space Present Position: Associate Professor, Penn-State University
2009/9 - 2013/8	Erickcek, Adrienne (Completed) , Perimeter Institute for Theoretical Physics/Canadian Principal Supervisor Institute for Theoretical Astrophysics Thesis/Project Title: Catastrophic Consequences of Kicking the Chameleon Present Position: Assistant Professor, University of North Carolina-Chapel Hill

Research Associate

2021/9 - 2022/5	Mukherjee, Suvodip (Completed) , Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background Present Position: Reader, Tata Institute of Fundamental Research, India
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Staff Supervision

Number of Visiting Researchers: 1

Event Administration

2020/7 - 2024/12	Member of the Organizing Committee, "Copernicus Weekly Webinar Series" focusing on cosmology and high energy physics https://indico.cern.ch/event/936284/overview , Seminar, 2020/7 - 2024/12
2023/3 - 2024/7	Local Organizing Committee (Chair), "50 Years of Horndeski Gravity: Exploring Modified Gravity" University of Waterloo and Perimeter Institute for Theoretical Physics, Conference, 2024/7 - 2024/7
2019/10 - 2024/6	Scientific Advisory Committee member, "[RN50] The Event Horizon and Beyond - Celebrating 50 Years of Narayan" Boston Museum of Science, Boston, MA https://www.cfa.harvard.edu/events/rn50-celebrating-50-years-narayan , Conference, 2024/6 - 2024/6
2023/11 - 2024/5	Local Organizing Committee (chair), "McNamara@65: Understanding Feedback in Galaxies and Clusters" University of Waterloo, Conference, 2024/5 - 2024/5
2023/11 - 2024/4	Scientific Organizing Committee Member, "Cosmological Frontiers in Fundamental Physics" (University of Edinburgh), Conference, 2024/4 - 2024/4
2022/10 - 2023/5	Organizing Committee (Chair), "Quantum Spacetime in the Cosmos: From Conception to Reality" Perimeter Institute, Conference, 2023/5 - 2023/5
2020/8 - 2021/5	Scientific Organizing Committee Member, "Cosmological Frontiers in Fundamental Physics" (Virtual at APC, Paris), Conference, 2021/5 - 2021/5
2020/2 - 2020/2	Organizer, "Echoes in Southern Ontario" Perimeter Institute, Workshop, 2020/2 - 2020/2

2019/5 - 2019/11	Co-organizer, Everpresent Lambda: Theory meets Observations, Conference, 2019/11 - 2019/11
2019/3 - 2019/9	Lead Organizer, "Cosmological Frontiers in Fundamental Physics 2019" Perimeter Institute, Conference, 2019/9 - 2019/9
2019/2 - 2019/7	Chairperson of Parallel Session, 22nd International Conference on General Relativity and Gravitation, Conference, 2019/7 - 2019/7
2017/9 - 2019/4	Chair, Astrophysics Seminar Chair, University of Waterloo, Seminar, 2017/9 - 2019/4
2019/1 - 2019/2	Scientific Organizing Committee, Tehran Meeting on Cosmology at the Crossroads (Virtual), Conference, 2019/2 - 2019/2
2017/9 - 2018/12	Chair, Physics 10 Undergraduate Lectures, University of Waterloo, Course, 2017/9 - 2018/12
2018/1 - 2018/6	Co-organizer, Asymptotic Safety in a Dark Universe, Conference, 2018/6 - 2018/6
2017/5 - 2017/11	Co-organizer, Quantum Black Holes in the Sky?, Conference, 2017/11 - 2017/11
2016/1 - 2016/7	Co-organizer, Philosophy of Cosmology, Conference, 2016/7 - 2016/7
2015/6 - 2015/12	Co-organizer, Mann Fest, Conference, 2015/12 - 2015/12
2015/2 - 2015/8	Co-organizer, Cosmic Flows (and Other Novelties on Large Scales), Conference, 2015/8 - 2015/8
2014/3 - 2014/4	Co-organizer, Implications of BICEP2, Workshop, 2014/4 - 2014/4
2011/5 - 2012/5	Co-organizer, Conformal Nature of the Universe, Conference, 2012/5 - 2012/5
2011/10 - 2012/4	Co-organizer, Theoretical Physics Workshop, Workshop, 2012/4 - 2012/4
2010/9 - 2011/9	Co-organizer, Unravelling Dark Matter 2011, Conference, 2011/9 - 2011/9
2009/12 - 2010/6	Co-organizer, Cosmological Frontiers of Fundamental Physics, Workshop, 2010/6 - 2010/6
2008/9 - 2009/8	Chair, Cosmology Seminar Series, Seminar, 2008/9 - 2009/8
2008/11 - 2009/5	Co-organizer, New Prospects for Solving the Cosmological Constant Problem, Workshop, - 2009/5
2009/5	
2008/5 - 2009/5	Co-organizer, Sunyaev-Zel'dovich Universe and the Future of Cluster Cosmology, Conference, 2009/4 - 2009/5
2008/10 - 2008/12	Co-organizer, Young Researchers Conference, Conference, 2008/12 - 2008/12
2008/4 - 2008/10	Co-organizer, PI/CITA Cosmology Mini-Workshop, Workshop, 2008/10 - 2008/10
2007/12 - 2008/6	Co-organizer, Small Scale Structure of Dark Matter, Workshop, 2008/6 - 2008/6
2008/1 - 2008/3	Co-organizer, Origins and Observations of Primordial Non-Gaussianity, Workshop, 2008/3 - 2008/3

Editorial Activities

2014/1 - 2020/8	Scientific Editor, Research in Astronomy and Astrophysics, Journal
2012/9 - 2013/8	Editorial Board Member, Phys13 News, Newsletter
2010/9 - 2011/8	Editorial Board Member, Phys13 News, Newsletter

Graduate Examination Activities

2024/9 - 2024/9	PhD External Examiner, Danieli, Valentina, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati
2024/9 - 2024/9	PhD External Examiner, Velluci, Vania, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati

Community and Volunteer Activities

2009/9	Speaker/Educator, Perimeter Institute for Theoretical Physics I am a regular contributor to outreach events at Perimeter Institute. My responsibilities include giving lectures, participating in Q&A's, and discussions with primary and high school students, as well as high school teachers.
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Knowledge and Technology Translation

2020/2 - 2023/5	Creator, Research Uptake Strategies Target Stakeholder: General Public Outcome / Deliverable: Based on our study of North American demographic, mobility, climate, and COVID-19 data, we provided a predictive model for growth and mitigation strategies for each US and Canadian city and county. The models were publicly available in the form of an online portal, during April 2021- May 2023. References / Citations / Web Sites: Canada: http://mylocalcovid.uwaterloo.ca [decommissioned in May 2023] US: https://wolfr.am/COVID19Dash
2020/4 - 2021/10	Scientific Advisor, Consultation Service Group/Organization/Business Serviced: Public Broadcasting Company (PBS) Target Stakeholder: General Public Outcome / Deliverable: NOVA Universe Revealed: Big Bang, Venture back to the birth of space and time as NOVA probes the mystery of the Big Bang. Premiered: 11/24/21 at 9pm EST Runtime: 53:16, PBS documentary Evidence of Uptake/Impact: While the specific viewership for this program is not publicly released, PBS is the premier public national broadcaster in the US. Each month, approximately 100 million people watch their local PBS stations. References / Citations / Web Sites: https://www.pbs.org/wgbh/nova/video/nova-universerevealed-big-bang/ Activity Description: I acted as the scientific advisor for the PBS NOVA documentary: "NOVA Universe Revealed: Big Bang", and helped with creating a scientifically accurate portrait of the Big Bang

International Collaboration Activities

2019/8 - 2029/8	Member of Laser Interferometer Space Antenna (LISA) consortium, Canada The Laser Interferometer Space Antenna (LISA) is a planned space-based observatory designed to detect and study gravitational waves. As a member of the LISA consortium, I am contributing to the agenda and ideas for the "Fundamental Physics" and "Waveform Modelling" working groups.
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Committee Memberships

2024/7 - 2025/7	Committee Member, Seminars and Colloquia Committee, Perimeter Institute for Theoretical Physics
2020/9 - 2024/8	Committee Member, Undergraduate Experience Committee, University of Waterloo Organize and coordinate events to improve student experience
2023/9 - 2024/7	Committee Member, Postdoctoral Search Committee, Perimeter Institute for Theoretical Physics
2021/9 - 2023/8	Chair, PhD Student Student Affairs, Perimeter Institute for Theoretical Physics
2013/1 - 2022/7	Committee Member, Early Researcher Award Grant Review Panel, Ministry of Research and Innovation, Government of Ontario I served on the review panel for most years, during this period.
2019/9 - 2020/8	Committee Member, Emmy Noether Fellowship Search Committee, Perimeter Institute for Theoretical Physics
2019/7 - 2019/10	Chair, "Astrophysics Theory Panel" grant review committee, National Aeronautics and Space Administration (NASA) Chair of a review panel for Theoretical Astrophysics grants
2014/8 - 2017/8	Chair, Colloquium Committee, Department of Physics and Astronomy, University of Waterloo
2013/9 - 2014/8	Committee Member, Co-op Report Marking Committee, University of Waterloo Ongoing Membership
2010/9 - 2012/8	Committee Member, Seminar/Colloquium Committee, University of Waterloo
2010/9 - 2011/8	Committee Member, SIN Exam Committee, University of Waterloo
2010/9 - 2011/8	Committee Member, PI/UW Astronomy Faculty Search Committee, University of Waterloo
2008/9 - 2010/8	Committee Member, Postdoctoral Recruitment Committee, Perimeter Institute for Theoretical Physics

Other Memberships

2021/1	Member, Canadian Institute for Theoretical Astrophysics, Incorporated (CITA, Inc) The Canadian Institute for Theoretical Astrophysics (CITA), located at the University of Toronto, is a leading research centre dedicated to exploring fundamental questions in theoretical astrophysics. CITA's research spans areas such as cosmology, galaxy formation, black holes, high-energy astrophysics, and planetary science. As a member of CITA Inc., I collaborate with other members across the country to advance theoretical astrophysics. As part of this organization, I contribute to recruiting top talent to Canada and mentoring highly qualified personnel, fostering the next generation of astrophysicists in this dynamic field.
2015/1	Ordinary Member, Canadian Astronomical Society (CASCA) CASCA is a professional society for Canadian astronomers and researchers involved in the fields of astronomy, astrophysics, and related disciplines. The primary goals of CASCA are to promote the advancement of astronomical research, education, and outreach in Canada.
2009/9 - 2014/8	Faculty of Science representative at the Faculty of Math Council, University of Waterloo Ongoing appointment

Presentations

- 1 (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Arizona State University, Phoenix, AZ, United States of America
- 2 (2025). Battle of the Big Bang: How the Universe Began (and Why It's a Big Mystery!). Physics Literacy for Kids (Zoom talk),
- 3 (2025). Copenhagen Survey on Black Holes and Fundamental Physics. Physics Colloquium (virtual), Tehran, Iran
- 4 (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics & Astronomy Colloquium, McMaster University, Hamilton, ON, Canada
- 5 (2025). Quantum Causality & the Crisis in Physics: Black Holes, the Big Bang, and Beyond. Sydney CPPC virtual seminar, Sydney, Australia
- 6 w/ Stefano Liberati (SISSA). (2024). Quantum Black Holes (concluding panel discussion, and wrap-up). Black Holes inside and out, Niels Bohr Institute, Copenhagen, Denmark
- 7 (2024). Precision Quantum Cosmology: Past, Present, and Future. Gravity and Cosmology 2024, Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan
- 8 (2024). Quantum Spacetime in the Sky: from the horizon to horizon. Heising-Simons Foundation (HSF) Collaboration on Quantum Gravity and Its Observational Signatures (QuRIOS) Journal Club (virtual), Waterloo, ON, Canada
- 9 (2024). Quantum Black Holes in the Sky. The Event Horizon and Beyond, Celebrating 50 years of Narayan, Boston, MA, United States of America
- 10 (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, International Center for Theoretical Physics Asia-Pacific (ICTP-AP), Beijing, China
- 11 (2024). Quantum Spacetime in the Sky: from the horizon to the vacuum. Bottom-Up Cross-Cutting Workshop "JENAS Initiative: Gravitational Wave Probes of Fundamental Physics" 12–16 Feb 2024 Sapienza University of Rome, Rome, Italy
- 12 (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Institute of High Energy Physics, Beijing, China
- 13 (2024). Quantum Spacetime in the Sky: from the horizon to horizon. International Congress of Basic Science 2, Beijing Institute for Mathematical Science and Applications (BIMSA), Beijing, China
- 14 (2024). Imaging Quantum Black Holes & (Towards) Optimal Estimators of Stochastic Gravitational Wave Background. The 1st Tsinghua Relativity Workshop, Beijing, China
- 15 (2023). The Big Picture: New Vistas in Gravitational Wave Astronomy. Canadian Astronomical Society (CASCAS) Annual General Meeting, Penticton, BC, Canada
- 16 (2023). Quantum Spacetime and the Cosmic Endgame. Physics Colloquium, Simon Fraser University, Burnaby, Canada
- 17 (2023). A Tale of Stars and Sirens. The Past, Present and Future of Astronomical Surveys - A Celebration of the Scientific Careers of Michael Strauss and Robert Lupton, Princeton University, Princeton, NJ, United States of America
- 18 (2023). Quantum Gravity and the Cosmos. Testing Aspects of General Relativity-II (virtual), Waterloo, ON, Canada
- 19 (2023). The paradox of human and the cosmos. Ottawa Intellectual Discussions (Virtual), Waterloo, ON, Canada
- 20 (2023). The Endgame: Quantum Spacetime and the Cosmos. Physics Colloquium, University of British Columbia, Vancouver, BC, Canada
- 21 (2023). The Big Picture. DE Shaw Company (virtual seminar), Waterloo, ON, Canada

- 22 (2023). Quantum Spacetime in the Sky: from Conception to Reality. ISRA 2023: The International Symposium on Recent Developments in Relativistic Astrophysics, Gangtok, India
- 23 (2023). Into the Unknown: A Journey on the Dark Side. Virtual Public Lecture for Lord Byng high school Astronomy Club (Vancouver, BC), Waterloo, ON, Canada
- 24 (2023). From Black Holes to the Big Bang: Humanity's Cosmic Paradox. Public Lecture at Kitchener Public Library, Kitchener, ON, Canada
- 25 (2023). Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada
- 26 (2023). Quantum Precision Cosmology. Testing Gravity 2023, Vancouver, BC, Canada
- 27 (2023). The War Within: Humanity's Cosmic Paradox. Physics 10 seminar series, University of Waterloo, Waterloo, ON, Canada
- 28 (2022). Einstein vs Einstein. Testing Aspects of General Relativity (Joint Canadian/Indian virtual conference), Waterloo, Canada
- 29 (2022). Einstein vs Einstein: The Great War. Physics Colloquium Institute for Research in Fundamental Sciences (IPM), (remote presentation), Tehran, Iran
- 30 (2022). Mapping the Big Bang. Remote Public Talk at Farzanegan High School, Tehran (virtual talk in Persian), Tehran, Iran
- 31 (2022). Stimulating Hawking Radiation of Gravitational waves: from theory to observation. New horizons for (no-)horizon physics: from gauge to gravity and back, Galileo Galilei Institute, Florence, Italy
- 32 (2022). Einstein vs Einstein: The Great War. Cosmo Coffee Seminar Series, Theory Group, the European Organization for Nuclear Research (CERN) (remote presentation), Geneva, Switzerland
- 33 (2022). Stimulating Hawking Radiation of Gravitational Waves. Quantum Simulators of Fundamental Physics (virtual seminar series), Waterloo, Canada
- 34 (2022). Quantum Black Holes: from holography to scrambling and gravitational wave astronomy. The 1st International Conference of Holography and its Applications, Damghan University (remote presentation), Damghan, Iran
- 35 (2022). Fundamental Physics vs. Fundamentalism. History, Philosophy, and Sociology of Cosmology and Astroparticle Physics Conference, Bonn, Germany
- 36 (2021). Cosmology of the Pandemic. Journal-club Galaxies, Institut d'Astrophysique de Paris (remote talk), Paris, France
- 37 (2021). Einstein vs Einstein: The Great War. Gravity Seminar Series, University of Southampton (remote presentation), Southampton, United Kingdom
- 38 (2021). Black Holes, Unscripted. Copernicus Webinar Series (virtual presentation), Waterloo, Canada
- 39 (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. Sixteenth Marcel Grossmann Meeting (virtual), Waterloo, ON, Canada
- 40 (2021). Physics and Astrophysics with Gravitational Waves. 14th Iranian national conference on Astronomy and Astrophysics, Semnan University (Virtual public talk in Persian), Semnan, Iran
- 41 (2021). Big Bang: Fact vs Fiction. The 28th IPM Physics Spring Conference, Tehran, Iran (remote presentation), Tehran, Iran
- 42 (2021). Cosmology and the Pandemic. Virtual conference on Gravity and Cosmology, Shahid Beheshti University (Remote keynote presentation in Persian), Tehran, Iran
- 43 (2021). A New Way to Measure Mass. Streams 21 (virtual conference), Waterloo, Canada

- 44 (2021). Cosmology: Triumphs, Trials, and Tribulations. Physics Colloquium, Tarbiat Modares University (remote presentation), Tehran, Iran
- 45 (2021). From Cosmos to the Pandemic. 37th Science and Technology Conference, Allameh Helli High School (virtual talk in Persian), Tehran, Iran
- 46 (2021). Into the Unknown. Physics Colloquium, University of Northern Iowa (remote), Cedar Falls, IA, United States of America
- 47 (2021). Black Holes, Unscripted. Black-Hole Microstructure, virtual conference organized by U-Saclay, France (remote presentation), Saclay, France
- 48 (2021). From Cosmology to the Pandemic. Sharif University of Technology Alumni Associate Webinar Series (virtual talk in Persian), Waterloo, ON, Canada
- 49 (2021). On Nothing!. Alternative Gravities and Fundamental Cosmology 2021 (virtual conference, organized by University of Szczecin, Poland), Szczecin, Poland
- 50 (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021, virtual), Waterloo, ON, Canada
- 51 (2021). From the Cosmos to the Pandemic: A Dive into the Unknown. 26th Special Physics School, Institute for Advanced Studies in Basic Sciences (remote talk in Persian), Zanjan, Iran
- 52 (2020). Probing Physics Beyond the Standard Model with the Gravitational Wave "Mystery" Noise. Quantum Spacetime and the Renormalization Group (organized by CP3 Origins, Denmark, virtual conference), Waterloo, ON, Canada
- 53 (2020). Local drivers of growth and containment for COVID-19 pandemic. Seminar Series on the Mathematical Modelling of COVID-19, Fields Institute for Research in Mathematical Sciences, Toronto, Canada
- 54 (2020). A New Way to Measure Mass. CANadian Virtual Astronomy Seminar (CANVAS) (virtual presentation), Waterloo, Canada
- 55 (2020). On the Edge of the Abyss. Virtual Seminar, McGill Space Institute, Montreal, QC, Canada
- 56 (2020). A New Way to Measure Mass. Galactic Dynamics Journal Club, University of Pennsylvania (remote presentation), Philadelphia, PA, United States of America
- 57 (2020). Quantum Black Hole Seismology. The 13th International LISA Symposium (virtual conference, prerecorded talk), Waterloo, Canada
- 58 (2020). Quantum Black Hole Seismology. 23rd meeting in Astronomical Research, Zanjan, Iran (remote presentation), Zanjan, Iran
- 59 (2020). A novel predictive framework for the novel coronavirus. Centre for Disease Modelling, Canada- China Distinguished Lecture, Canada
- 60 (2020). Quantum Black Holes in the Sky. Quantum Gravity and the Renormalization Group, Odense, Denmark
- 61 (2020). Managing the COVID-19 Pandemic across Geography and Demography. seminar for the Google COVID-19 data working group (virtual), Waterloo, ON, Canada
- 62 (2020). On the secrets of black holes, worm holes, white holes, and the firewalls. Remote public talk in Persian for Astronomy (Nojoom) Magazine, Tehran, Iran
- 63 (2020). Quantum Black Hole Seismology. Astrophysics Seminar Series, Bar-Ilan University, Israel (remote presentation), Israel
- 64 (2020). Quantum Black Hole Seismology. High Energy Physics (HEP) seminar series, school of physics, Institute for Research in Fundamental Sciences (IPM), Tehran (remote presentation), Tehran, Iran
- 65 (2020). Managing the COVID-19 Pandemic across Geography and Demography. Seminar Series, Astronomy School, Institute for Research in Fundamental Sciences (IPM), Tehran (remote, tehran, Iran

- 66 (2020). A New Way to Measure Mass. Astrophysics Seminar, Sharif University (remote presentation), Tehran, Iran
- 67 (2020). The invisible 99 percent. Physics Club (remote public talk in Persian), Tehran, Iran
- 68 (2020). Managing the COVID-19 Pandemic across Geography and Demography. Colloquium, Perimeter Institute, Waterloo, ON, Canada
- 69 (2020). Hubble Tension and an Everpresent Lambda. 3rd Online Workshop on Quantum Gravity, “Quantum Gravity meets dark energy” (virtual conference), Waterloo, Canada
- 70 *Aslanbeigi, S; *Saravani, M. (2019). Dark Matter from Quantum Gravity. Everpresent Lambda: Theory Meets Observations, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada
- 71 (2019). Dark Energy [On Nothing]. CosmoCruise 2019, Venice, Italy
- 72 (2019). A Tale of Big Bang. Milky Way Persian Science Channel, Iran (virtual), Waterloo, ON, Canada
- 73 (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, BC, Canada
- 74 (2019). Cosmological non-Constant (CnC) Problem. 5th Bahamas Advanced Study Institute and Conferences 2019 (BASIC 2019) (remote presentation), Bahamas
- 75 (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, Canada
- 76 (2019). On Nothing!. Physics Colloquium at Grand Valley State University, Grand Rapids, MI, United States of America
- 77 (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. Solvay-Perimeter-APC Workshop on "Cosmological Frontiers in Fundamental Physics" - Solvay Institute, Brussels, Belgium
- 78 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar Series, Perimeter Institute, Waterloo, ON, Canada
- 79 (2018). Seeing through Simulation Blindspots. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada
- 80 (2018). Reviving Aether in Einstein’s Universe: Trials, Tribulations, and Triumphs. Lecture for undergraduate Cosmology and Gravitation class at Western University, London, ON, Canada
- 81 (2018). Seeing Cosmic Nanostructure: From Plasma to Dark Matter. Seminar - Canadian Institute for Theoretical Astrophysics, University of Toronto, Toronto, Canada
- 82 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Physics and Astronomy Colloquium, Western University, London, ON, Canada
- 83 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Theoretical High Energy Physics Seminar Series, McGill University, Montreal, QC, Canada
- 84 (2018). Adios, Lorentz! Hola, Kuhn!. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada
- 85 (2018). Cosmological non-Constant Problem. Asymptotic Safety in a Dark Universe, Perimeter Institute, Waterloo, ON, Canada
- 86 (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. HEP Seminar - McGill University, Montreal, Canada
- 87 (2018). The Other 99 Percent: A Dark Cosmos. Keynote lecture for high school teachers, Perimeter Institute, Waterloo, ON, Canada
- 88 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada

- (2017). Echoes from the Abyss. Testing Gravity 2017, Vancouver, Canada
89
- (2017). Reflections on Spacetime. Cosmology and the Future of Spacetime - Western University, London, Canada
90
- (2017). Echoes from the Abyss. Seminar - Department of Applied Mathematics, University of Waterloo, Waterloo, Canada
91
- (2017). Quantum Black Holes in the Sky. Colloquium - University of Kansas, Lawrence, Kansas, United States of America
92
- (2017). Quantum Black Holes in the Sky. Colloquium - Department of Physics, Ohio State University, Columbus, United States of America
93
- (2017). Quantum Black Holes in the Sky - Plenary Lecture. CosPA 2017, Kyoto, Japan
94
- (2017). How to turn Quantum Gravity into Science?. Probing the Spacetime Fabric: From Concepts to Phenomenology - SISSA, Trieste, Italy
95
- (2017). Echoes from the Abyss. Physics and Astrophysics at the eXtreme (PAX), Amsterdam, Netherlands
96
- (2017). Seeing Dark in the Extremes: From Nanostructure, to Infinity and Beyond. CPARC Colloquium - Queen's University, Kingston, Canada
97
- (2016). The Many Faces of the Cosmological Constant Problem. Experimental Search for Quantum Gravity - Frankfurt Institute for Advanced Studies, Frankfurt, Germany
98
- (2016). Cosmological Non-Constant Problem: Astrophysical Limits of Particle Physics. HET-Discovery Seminar - Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark
99
- (2016). E.F.T. is Evil!. Unifying Tests of General Relativity - Walter Burke Institute for Theoretical Physics, Caltech, Pasadena, United States of America
100
- (2016). Accretion in Radiative Equipartition. Seminar - Dark Cosmology Centre, University of Copenhagen, Copenhagen, Denmark
101
- (2015). (Real) Cosmology with Quark-Gluon Plasmas. Cosmology/HEP Seminar - Brookhaven National Laboratory, Upton, New York, United States of America
102
- (2015). Pulsar Timing Constraints on Physics Beyond Standard Model. Mann Fest - University of Waterloo, Waterloo, Canada
103
- (2015). Cosmological (Non-)Constant Problem. Testing Gravity 2015, Vancouver, Canada
104
- (2015). First Erlangen Workshop on Quantum Gravity and Cosmology. High Energy Physics Seminar - University of Toronto, Toronto, Canada
105
- (2015). Cosmological Non-Constant Problem: The Case for TeV Scale Quantum Gravity. First Erlangen Workshop on Quantum Gravity and Cosmology, Erlangen, Germany
106
- (2015). Alice in the (Relativistic) Wonderland. Colloquium - Department of Mathematics, Western University, London, Canada
107
- (2014). Probing Dark Matter Nanostructure. Mitchell Workshop on Collider and Dark Matter Physics, College Station, United States of America
108
- (2014). From Black Holes to Big Bang, and Back!. Cosmology Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
109
- (2014). Cosmological (Non-)Constant Problem. Large Angle Anomalies: from a posteriori to a priori - Case-Western Reserve University, Cleveland, United States of America
110
- (2014). An Introduction to the Universe - Plenary Talk. Sharif University Alumni Conference, Tehran, Iran
111

- 112 (2014). From Black Holes to Big Bang, and Back!. Colloquium - Department of Physics, Isfahan University of Technology, Isfahan, Iran
- 113 (2014). Probing Dark Matter Nanostructure. Cosmology Seminar - Department of Physics, Sharif University of Technology, Tehran, Iran
- 114 (2014). Dark Matter Nanostructure. Astrophysics Seminar - Department of Physics and Astronomy, University of Waterloo, Waterloo, Canada
- 115 (2014). From Black Holes to Big Bang, and Back!. Invited Colloquium - School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
- 116 (2014). An Introduction to the Universe, Q&A. Zaferanieh Amateur Observatory, Tehran, Iran
- 117 (2014). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Physics Colloquium - University of Tehran, Tehran, Iran
- 118 (2013). Missing Baryons: How Much? How Hot? How Fast?. Cross-Correlating Cosmic Fields Conference, Shanghai, China
- 119 Gould E. (2013). A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics?. Quantum Foundations Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 120 (2013). Tales of Dark Matter on Small Scales. Colloquium - Kavli Institute for Astronomy and Astrophysics, Beijing, China
- 121 (2013). Dynamical Emergence of Universal Horizons During the Formation of Black Holes. LOOPS 13, Waterloo, Canada
- 122 (2013). Out of the White Hole: A Holographic Origin for the Big Bang. Cosmology After Planck 2013, Ann Arbor, United States of America
- 123 (2013). Where Will Einstein Fail?. CASCA 2013, Vancouver, Canada
- 124 (2013). Why is Particle Physics Lorentz-Invariant?. Kavli Workshop - IPMU Focus Week on Gravity and Lorentz Violation, Kashiwa, Japan
- 125 (2013). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Colloquium - University of Wisconsin-Madison, Madison, United States of America
- 126 (2013). Tales of Dark Matter on Small Scales. TAMU Workshop on Dark Matter, College Station, United States of America
- 127 (2012). Horava-Lifshitz Cosmology and Dark Matter Holes - 2-Lecture Series. Cape Town International Cosmology School 2012, Stellenbosch, South Africa, Republic of
- 128 (2012). Hot Big Bang at a Lifshitz Point. CAP Congress 2012, Calgary, Canada
- 129 (2012). Reviving Gravity's Aether in Einstein's Universe. Colloquium - York University, Toronto, Canada
- 130 (2012). Tales of Dark Matter on Small Scales. Theoretical Physics Workshop - UWO, London, Canada
- 131 (2012). Why is Particle Physics Lorentz Invariant?. Experimental Search for Quantum Gravity: The Hard Facts, Waterloo, Canada
- 132 (2012). Modern Cosmology. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 133 Bonanno A, Brandenberger R, Magueijo J, Nemiroff R, Amelino-Camelia G. (2012). Photons and Quantum Gravity in Astrophysics. Experimental Search for Quantum Gravity 2012, Waterloo, Canada
- 134 (2012). Trials and Tribulations of Aether. Conformal Nature of the Universe Conference, Waterloo, Canada
- 135 (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. CITA Invited Seminar - University of Toronto, Toronto, Canada

- 136 (2011). Largest and Smallest Dark Matter Clusters, and What We Can Learn from Them. High Energy Theory
Seminar - University of Michigan, Ann Arbor, United States of America
- 137 (2011). Structure in the Phase Space and Dark Matter Astronomy. Unravelling Dark Matter Conference 2011,
Waterloo, Canada
- 138 (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. KICP Colloquium - University of
Chicago, Chicago, United States of America
- 139 (2011). Aether, Quantum Gravity and Cosmology. Philosophy of Cosmology Workshop - 11th Annual
Conference in the Foundations of Physics, London, Canada
- 140 (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. Vainu Bappu Memorial Gold Medal
Award Ceremony - Astronomical Society of India, Pune, India
- 141 (2011). Aether, Quantum Gravity and Cosmology. Cosmological Frontiers in Fundamental Physics 2011, Paris,
France
- 142 (2011). How can Astrophysical Compact Objects Shed Light on Dark Cosmology?. High Energy Theory
Seminar - University of Amsterdam, Amsterdam, Netherlands
- 143 (2011). Fine Structure of Dark Matter Phase Space. Dark Matter: Its Origin, Models, and Detection,
Albuquerque, United States of America
- 144 (2010). Non-Linear Structure Formation in Cosmology - 2-Part Lecture Series. Invited Lectures on Structure
Formation - Institute for Research in Fundamental Sciences, Tehran, Iran
- 145 (2010). First Precision Measurement in Quantum Gravity. Cosmology Seminar - Institute for Research in
Fundamental Sciences, Tehran, Iran
- 146 (2010). Stellar Black Holes and the Thermodynamic Origin of Cosmic Acceleration. Colloquium - Institut
d'Astrophysique de Paris, Paris, France
- 147 (2010). Cosmological Constant Problem: Rethinking Quantum and Gravity. Laws of Nature: Their Nature and
Knowability 2010, Waterloo, Canada
- 148 (2010). Cosmology and the End of Physics!. Colloquium - University of Guelph, Guelph, Canada
- 149 (2010). Hierarchy in the Phase Space and Dark Matter Astronomy. Astrophysics Seminar - Queen's University,
Kingston, Canada
- 150 (2009). The End of the Cosmological Constant Problem. Astrophysics Workshop - Interactions in the Dark
Sector, Leiden, Netherlands
- 151 (2009). Black Hole Horizons and the Origin of Cosmic Acceleration. Holographic Cosmology Workshop 2009,
Waterloo, Canada
- 152 (2009). Gravitational Aether as a Solution to the CC Problem and more!. New Prospects for Solving the
Cosmological Constant Problem 2009, Waterloo, Canada
- 153 (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Cosmology Seminar - KIPAC, Stanford,
United States of America
- 154 (2009). Hierarchy in the Phase Space and Dark Matter Astronomy. CYGNUS 2009, Cambridge, United States
of America
- 155 (2009). The End of the Cosmological Constant Problem. Cosmology Seminar - IPMU, Kashiwa, Japan
- 156 (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Theory Canada 5 Conference, Fredericton,
Canada
- 157 (2009). Cuscuton and Solving the Cosmological Constant Problem. High Energy Physics Seminar - Vanderbilt
University, Nashville, United States of America
- 158 (2009). Cuscuton and Horava-Lifshitz Gravity. Gravity at a Lifshitz Point 2009, Waterloo, Canada

- 159 (2009). (Towards) the End of the Cosmological Constant Problem!. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 160 (2008). Gravitational Aether as a Solution to the Cosmological Constant Problem. 14th International Symposium on Particles, Strings and Cosmology (PASCOS '08), Waterloo, Canada
- 161 (2008). Physics Beyond the Horizon. Cosmology and Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 162 (2008). Physics Beyond the Horizon. Physics Colloquium - University of Pennsylvania, Philadelphia, United States of America
- 163 (2008). Physics Beyond the Horizon. Colloquium - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 164 (2008). Observational Evidence for Massive Gravity. PI CITA Joint Cosmology Day 2008, Waterloo, Canada
- 165 (2008). Missing Thermal Energy of the Universe. Astroparticle Workshop - Open Questions from Clusters of Galaxies, Toulouse, France
- 166 (2008). Primordial Non-Gaussianity, Statistics of Collapsed Objects, and the ISW Effect. Origins and Observations of Primordial Non-Gaussianity 2008, Waterloo, Canada
- 167 (2008). Galaxy Clusters as Cosmic Canaries. High Energy Physics Seminar - Syracuse University, Syracuse, United States of America

Broadcast Interviews

- 2024/09/27 - Physicists search for Black Hole Bombs, Phil Halper's YouTube Channel, YouTube.com, <https://youtu.be/bJlqWGgt2hw?si=cn9waXQiKvUHIZw3>
- 2024/10/12 - What's inside a black hole? Fuzzballs, Echoes & The Big Bang, Phil Halper's YouTube Channel, YouTube.com, <https://youtu.be/JN9jlg9AdYM?si=c4158UQDt8pRFAfn>
- 2024/05/17 - Did the Universe Create itself ? The PTC model, Before the Big Bang 11, YouTube.com, <https://youtu.be/cLSuM8rKMyk?si=x dh2appUW8wAEi-7>
- 2024/10/12 - The Fine Tuning Argument: the critics strike back, Phil Halper's YouTube channel, YouTube.com, https://youtu.be/zNH-ZgSpBuQ?si=fahqxSe3_F4xgzDv
- 2022/10/09 - Echoes from the Abyss: The Mind-Bending Science Behind the Strange Sounds of Black Holes, Time before Space, YouTube.com https://www.youtube.com/watch?v=d1mcfBip_sw&ab_channel=TimebeforeSpace
- 2024/10/12 - Ghazal Geshnizjani & Niayesh Afshordi: Cosmologically Coupled, Conversations at Perimeter Institute, <https://conversationsattheperimeter.podbean.com/>
- 2020/10/28 - A New Model for the Covid Pandemic, Sabine Hossenfelder's channel, YouTube.com, <https://youtu.be/H0oZUpMtJrs?si=TA5cEInNhZRnN5ug>
- 2024/10/12 - Astronomers claim universe has "cosmic glitch", Phil Halper's Youtube Channel <https://>
- 2024/05/03

2024/05/03	youtu.be/GOctev5cUCI?si=LacFQqSuTPKsAUyO, Youtube
2020/11/20	- Fighting COVID-19 with Physics, Interview with the local CTV station in Kitchener-Waterloo, CTV News, Kitchener-Waterloo
2021/11/20	- Q&A with the Experts: The problem with Herd Immunity and COVID-19, Mike Farwell Show, 570 News, Kitchener Radio
2020/11/12	- Black Hole Genesis, Before the Big Bang, Episode 10, YouTube.com 2019/10/04
2019/10/04	- Varying Speed of Light Cosmology, Before the Big Bang, Episode 8, YouTube.com 2019/07/31
2018/10/17	- Big Bang and Holography, Science Friday with Ira Flatow, National Public Radio, USA 2017/02/10
2017/02/10	- Big Bang, Big Think, Radio Canada International (https://www.rcinet.ca/en/2017/02/06/the-universe-is-holographic/)
2017/02/06	- Holographic Cosmology, Morning Edition, CBC Kitchener-Waterloo 2017/02/01
2017/02/01	- Enigmatic echo from Black Hole, Science@ORF.at, Austrian Broadcasting Corporation 2017/01/01
2017/01/01	- Bye Bye Big Bang?, FQXi Podcast, FQXi.org 2013/10/13
2013/10/13	- Working at PI, Fresh Air with Mary Ito, CBC Radio 2013/03/16
2013/03/16	- Discovery of Higgs Particle at CERN, 60 Minutes, BBC Persian TV 2012/07/04
2012/07/04	- Legacy of Stephen Hawking, 60 Minutes, BBC Persian TV 2011/10/04
2011/10/04	- (Claimed) Discovery of Superluminal Neutrinos, 60 Minutes, BBC Persian TV 2011/09/24
2011/09/24	

Text Interviews

2024/05/15	Newfound 'glitch' in Einstein's relativity could rewrite the rules of the universe, study suggests, LiveScience.com https://www.livescience.com/physics-mathematics/newfoundglitch-in-einsteins-relativity-could-rewrite-the-rules-of-the-universe-study-suggests
2024/05/15	A Cosmic Glitch in Gravity, Nautilus Magazine https://nautil.us/a-cosmic-glitch-in-gravity-593171/?_sp=0f3668e3-5102-4de7-b03f-fc59eb4ec411.1715821798622
2024/05/01	A 'cosmic glitch' in gravity, Waterloo News https://uwaterloo.ca/news/media/cosmic-glitchgravity
2024/01/29	Over Gravity's Rainbow: An alternative framework of physics, in which light's speed can vary By Zeeya Merali, John Templeton Foundation Newsletter https://www.templeton.org/news/over-gravitys-rainbow
2023/02/17	"Dark energy from supermassive black holes? Physicists spar over radical idea", Science News https://www.science.org/content/article/dark-energy-supermassive-black-holesphysicists-spar-over-radical-idea

2021/09/20 Chiamaka Okoli was a rarity in physics. She challenged norms until her untimely death A Black woman, mother, and a devout Catholic, Okoli was a trailblazer in her field before tragedy struck, Boradview.com <https://broadview.org/chiamaka-okoli/>

2021/06/04 Modelling the mysteries of the universe, closer to home Professor Niayesh Afshordi used his knowledge of modelling the mysteries of the Cosmos to start modelling the mysteries of COVID-19, Waterloo News <https://uwaterloo.ca/news/science/modeling-mysteriesuniverse-closer-home>

2021/02/08 Echoes from black hole could force us to rethink laws of physics Gravitational echoes may confirm the late Stephen Hawking’s quantum black hole hypothesis, Waterloo News, <https://uwaterloo.ca/news/global-impact/echoes-black-hole-could-force-us-rethink-lawsphysics>

2020/02/10 Spacetime ‘Echoes’ From Quantum Black Holes Could Soon Change Physics Forever, VICE Motherboard

2020/02/01 Musta aukko sateilee, TAHDET JA AVARUUS Finland

2020/01/31 Black holes shouldn’t echo, but this one might. Score 1 for Stephen Hawking?, Live Science

2019/10/30 Einstein killed the aether. Now the idea is back to save relativity, New Scientist

2019/06/18 Big Bang BOMBSHELL: Are we living inside a HUGE higher dimensional black hole?, Express.co.uk, <https://www.express.co.uk/news/science/1140059/black-hole-higherdimensional-big-bang-theory-universe-hologram-space-news>

2019/05/24 We’ve Found Gravitational Waves. What Will We Learn From Them?, Discover Magazine

2019/05/15 Ripple Effect, Discover Magazine

2019/01/16 Crisis at the Heart of Physics, New Scientist

2018/06/20 Total Wipeout, New Scientist

2018/05/14 Echoes of gravitational waves could point to quantum gravity, Physics World

2017/03/29 Echoes of a Black Hole, Aeon.COM

2017/02/04 Could the universe actually be a flat hologram?, Christian Science Monitor

2017/01/31 Lost in space? A brief guide to the ‘holographic principle’ of the universe, Guardian

2016/12/13 Was Einstein wrong? Controversial theory claims the speed of light is not a constant, Wired Magazine

2016/11/28 Theory Challenging Einstein’s View On Speed of Light Could Soon Be Tested, The Guardian

2016/09/12 LIGO black hole echoes hint at general-relativity breakdown, Nature News

2016/02/05 Much Ado around Nothing: The Cosmological non-Constant Problem, Backreaction

2016/01/11 The Sky as a Limit, Perimeter Institute News

2015/02/11 Do Black Hole Firewalls Have Observable Consequences?, Backreaction

2013/09/26 Was Our Universe Born in an Extra-Dimensional Black Hole?, PBS NOVA

2013/09/13	Did a Hyper-Black Hole Spawn the Universe?, Nature News
2013/07/16	Cosmologist Claims Universe May Not Be Expanding, Nature News
2010/07/01	Cosmic Bubble Made Cold Spot in Big Bang Afterglow, New Scientist
2009/03/16	Gravity May Venture Where Matter Fears to Tread, New Scientist

Publications

Journal Articles

- 1 Friedman-Shaw, B*; Krolewski, A*; Foglieni, M; Afshordi, N. (2025). Doppler Bias: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys. *Journal of Cosmology and Astroparticle Physics* (arXiv preprint: 2410.04705). 2025 Published, Institute of Physics,
- 2 Liu, R*; Quintin, J*; Afshordi, N. (2025). Conformal cores of quantum black holes in quadratic gravity. *Physical Review D*. 111(4): 044031. <http://dx.doi.org/https://doi.org/10.1103/PhysRevD.111.044031> Published, American Physical Society,
- 3 Tang, R*; Faraji, S*; Afshordi, N. (2025). Accretion disk dynamics in teleparallel Born-Infeld gravity. *Journal of Cosmology and Astroparticle Physics* (arXiv:2503.15776).
- 4 Chen, A*; Afshordi, N. (2025). Where are all the dark galaxies? Predicting galaxy/halo locations from their bright neighbors. *Physical review D* (arXiv preprint arXiv:2507.01814).
- 5 Berti, Emanuele and Cardoso, Vitor and Carullo, Gregorio and Abedi, Jahed and Afshordi, Niayesh and Albanesi, Simone and Baibhav, Vishal and Bhagwat, Swetha and Bl{\`a}zquez-Salcedo, Jos{\`e} Luis and Bonga, B{\`e}atrice and others. (2025). Black hole spectroscopy: from theory to experiment. arXiv preprint arXiv:2505.23895.
- 6 Goel, L*; Patterson, E A*; Preciado-Rivas, M R; Torabian, M; Mann, R B; Afshordi, N. (2025). Accelerated detector in a superposed spacetime. *Physical Review D* (arXiv preprint arXiv:2409.06818). 111(2): 025015. <http://dx.doi.org/https://doi.org/10.1103/PhysRevD.111.025015> Published, American Physical Society,
- 7 R{\`a}dler, Martin and Afshordi, Niayesh and Taleei, Reza and Parodi, Katia and Abolfath, Ramin and Lascaud, Julie. (2025). A semiclassical model of the immediate temperature distribution surrounding the track of heavy ions with therapeutic energies. *Physics in Medicine & Biology*. 70(11): 115010. , IOP Publishing,
- 8 Chen, A Y*; Halper, P; Afshordi, N. (2025). Copenhagen Survey on Black Holes and Fundamental Physics. *Studies in History and Philosophy of Physics* (arXiv:2503.15776).
- 9 Yang, J*; Afshordi, N; Torabian, M; Jafari, SA; Baskaran, G. (2025). Smart Holes: Analogue black holes with the right temperature and entropy. *Journal of High Energy Physics, JHEP07(2025)226*, (arXiv preprint 2412.08517). 2025(226)
- 10 Abolfath, R; Afshordi, N; Rahvar, S; van Duin, A; Taleei, R; Parodi, K; Lascaud, J; Mohan, R; et al. (2024). A molecular dynamics simulation of the abrupt changes in the thermodynamic properties of water after formation of nano-bubbles/nano-cavities induced by passage of charged particles. *The European Physical Journal D* (arXiv preprint arXiv:2403.05880). 78(11) Published, Springer Nature,
- 11 Mylova, M*; Afshordi, N. (2024). Effective cuscuton theory. *Journal of High Energy Physics*. 2024(4): 1–20. Published, Springer Berlin Heidelberg,
- 12 Wen, R Y*; Hergt, L T; Afshordi, N; Scott, D. (2024). A cosmic glitch in gravity. *Journal of Cosmology and Astroparticle Physics*. 2024(03): 045. Published, IOP Publishing,
- 13 Sajadian, S; Afshordi, N. (2024). Simulating Self-Lensing and Eclipsing Signals due to Detached Compact Objects in the TESS Light Curves. *Astronomical Journal* (arXiv preprint arXiv:2409.12441). 168(6): 298. Published, The American Astronomical Society,
- 14 Li, Y*; Liu, R*; Dailey, C*; Afshordi, N. (2024). Detecting cosmological scalar fields using orbital networks of quantum sensors. *Physical Review D*, 110 (4): 043028 (arXiv preprint arXiv:2311.17873).
- 15 Dey, R*; Micchi Longo, L F*; Mukherjee, S; Afshordi, N. (2024). Spectrogram correlated stacking: A novel time-frequency domain analysis of the stochastic gravitational wave background. *Physical Review D*. 109(2): 023029. Published, American Physical Society,
- 16 *Dailey, C; Schnetter, E; Afshordi, N. (2024). Formulating the complete initial boundary value problem in numerical relativity to model black hole echoes. *Classical and Quantum Gravity* (arXiv preprint arXiv:2409.17970).

- 17 Chen, A Y*; Afshordi, N. (2023). Outskirts of dark matter haloes. *Physical Review D*. 107(10): 103526. Published, American Physical Society,
- 18 Dailey, C*; Afshordi, N; Schnetter, E. (2023). Reflecting boundary conditions in numerical relativity as a model for black hole echoes. *Classical and Quantum Gravity*. 40(19): 195007. Published, IOP Publishing,
- 19 Borissova, J N*; Held, A*; Afshordi, N. (2023). Scale-invariance at the core of quantum black holes. *Classical and Quantum Gravity*. 40(7): 075011. Published, IOP Publishing,
- 20 Abedi, J*; Longo Micchi, L. F.*; Afshordi, N. (2023). GW190521: Search for echoes due to stimulated hawking radiation from black holes. *Physical Review D*. 108(4): 044047. Published, American Physical Society,
- 21 Oshita, N*; Afshordi, N. (2023). From entropy to echoes: Counting the quasi-normal modes and the quantum limit of silence. *Physics Letters B*. 841: 137901. Published, North-Holland,
- 22 Wu, D; Gao, P; Ren, J; Afshordi, N. (2023). Model-independent search for the quasinormal modes of gravitational wave echoes. *Physical Review D*. 108(12): 124006. Published, American Physical Society,
- 23 Afshordi, N; Magueijo, J. (2022). Lower bound on the cosmological constant from the classicality of the early Universe. *Physical Review D*. 106(12): 123518. Published, American Physical Society,
- 24 Yang, T*; Hudson, MJ; Afshordi, N. (2022). A universal profile for stacked filaments from cold dark matter simulations. *Monthly Notices of the Royal Astronomical Society*. 516(4): 6041–6054. Published, Oxford University Press,
- 25 Mylova, M*; Moschou, M; Afshordi, N; Magueijo, Jo. (2022). Non-Gaussian signatures of a thermal Big Bang. *Journal of Cosmology and Astroparticle Physics*. 2022(07): 005. Published, IOP Publishing,
- 26 Saraswat, K*; Afshordi, N. (2022). Spacing statistics of energy spectra: random matrices, black hole thermalization, and echoes. *Journal of High Energy Physics*. 2022(4): 1–58. Published, Springer Berlin Heidelberg,
- 27 Abolfath, R; Baikalov, A; Bartsch, S; Afshordi, N; Mohan, R. (2022). The effect of non-ionizing excitations on the diffusion of ion species and inter-track correlations in FLASH ultra-high dose rate radiotherapy. *Physics in Medicine & Biology*. 67(10): 105005. Published, IOP Publishing,
- 28 Arun, KG; et al. (2022). New horizons for fundamental physics with LISA. *Living Reviews in Relativity*. 25(1): 4. Published, Springer International Publishing Cham,
- 29 Conklin, R*; Afshordi, N. (2021). Boltzmann meets Lorentz: A surrogate model for black hole echoes. *arXiv preprint arXiv:2201.00027*.
- 30 Chua, WZ*; Afshordi, N. (2021). Electromagnetic albedo of Quantum Black Holes. (arXiv:2103.05790) *Journal of High Energy Physics*. 2021(7): 6.
- 31 Oshita, N*; Afshordi, N; Mukohyama, S. (2021). Lifshitz scaling, ringing black holes, and superradiance. *Journal of Cosmology and Astroparticle Physics*. 2021(05): 005. Published, IOP Publishing,
- 32 Longo Micchi, LF*; Afshordi, N.; Chirenti, C. (2021). How loud are echoes from exotic compact objects?. (arXiv:2010.14578) *Physical Review D*. 103(4): 044028.
- 33 Saraswat, K* Afshordi, N. (2021). Extracting Hawking radiation near the horizon of AdS black holes. (arXiv:2003.12676) *Journal of High Energy Physics*. 2021(2): 77.
- 34 Adshead, P Afshordi, N Dimastrogiovanni, E Fasiello, M Lim, E. A. Tasinato, G. (2021). Multimessenger cosmology: Correlating cosmic microwave background and stochastic gravitational wave background measurements. (arXiv:2004.06619) *Physical Review D*. 103(2): 023532.
- 35 Robbins, MPG*; Afshordi, N; Jamison, AO; Mann, RB. (2021). Detection of Gravitational Waves using Parametric Resonance in Bose-Einstein Condensates. (arXiv:2101.03691) *Journal of Cosmology and Astroparticle Physics*. 2101.03691: 1-20.
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- 37 Abedi, J*; Afshordi, N. (2020). Echoes from the Abyss: A Status Update. (arXiv:2001.00821) Preprint. 2001.00821: 1-8.

- 38 Oshita, N* Tsuna, D Afshordi, N. (2020). Quantum Black Hole Seismology II: Applications to Astrophysical Black Holes. (arXiv:2004.06276) Physical Review D. 102(2): 024046.
- 39 Yang, T* Hudson, M Afshordi, N. (2020). How dark are filaments in the cosmic web?. (arXiv:2001.10943) Monthly Notices of Royal Astronomical Society. 498(3): 3158-3170.
- 40 Wang Q*, Oshita N*, Afshordi N. (2020). Echoes from Quantum Black Holes. (arXiv:1905.00446) Physical Review D 101, 024031 (2020). 101(2): 024031.
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Books

Afshordi, Niayesh; Halper, Philip. (2025). Battle of the Big Bang: The New Tales of Our Cosmic Origins. 1:

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- 360 Calamia, Joseph. Co-Author Published, University of Chicagao Press, Description / Contribution Value: A thrilling exploration of competing cosmological origin stories, comparing new scientific ideas that upend our very notions of space, time, and reality. By most popular accounts, the universe started with a bang some 13.8 billion years ago. But what happened before the Big Bang? And how do we know it happened at all? Here prominent cosmologist Niayesh Afshordi and science communicator Phil Halper offer a tour of the peculiar possibilities: bouncing and cyclic universes, time loops, creations from nothing, multiverses, black hole births, string theories, and holograms. Along the way, they offer both a call for new physics and a riveting story of scientific debate.

Book Chapters

- *Altamirano, N; *Gould, E; Afshordi, N; Mann, RB. (2025). Cosmological Perturbations in the 5D Holographic Big Bang Model. Dr. Piero Nicolini, Ph.D. Touring the Planck Scale: Antonio Aurilia Memorial Volume (arXiv Preprint: 1703.00954). 1(1): 1-21. In Press, Springer Cham,

Supervised Student Publications

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Magazine Entries

- Afshordi N, Mann R, Pourhasan R*. The Black Hole That Birthed the Big Bang. Scientific American. 311(2)
1 Number of Contributors: 3 Description / Contribution Value: The article, which was featured on the cover of this issue, describes one of our most innovative ideas to a popular audience.

Online Resources

- Co-Author. Zavala, J, Afshordi, N. P2SAD: Particle Phase Space Average Density (public code). (2014).
1 Number of Contributors: 2 Description / Contribution Value: P2SAD computes the Particle Phase Space Average Density (P2SAD) in galactic haloes. The model for the calculation is based on the stable clustering hypothesis in phase space, the spherical collapse model, and tidal disruption of substructures. The multiscale prediction for P2SAD computed by this IDL code can be used to estimate signals sensitive to the small scale structure of dark matter distributions (e.g. dark matter annihilation). The code computes P2SAD averaged over the whole virialized region of a Milky-Way-size halo at redshift zero.

Conference Publications

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- *Wen, RY; Hergt, LT; Afshordi, N; Scott, D. (2024). A glitch in gravity: cosmic Lorentz-violation from fiery
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- Afshordi, N Holder, B Bahrami, M Lichtblau, D. (2021). Diverse local epidemics reveal the distinct effects of
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Talks and outreach materials archive

Complete inventory of 570 source decks and slide exports in the maintained local archive, catalogued on 2026-07-09. Browse the web collection at nafshordi.com/talks.

2026 — 8 items

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- Battle of the Big Bang KPL 2026
Keynote | public talk
- Hangzhou panel Afshordi
Keynote | Quadratic Gravity
- Hangzhou panel Afshordi
PDF | Quadratic Gravity
- neutrino PI 26
Keynote | massive neutrinos
- QG Wild Hangzhou 2026
Keynote | Quadratic Gravity
- QG Wild Hangzhou 2026
PDF | Quadratic Gravity
- QQG SAB
Keynote | Quadratic Gravity

2025 — 26 items

- Afshordi QBH Simons 2025
Keynote | Quantum Black Holes
- Battle of the Big Bang Secular AZ
Keynote | public talk
- CnC PI cosmo group 2025
Keynote | CNC
- Cooling+Heating McNamara@65
Keynote | DM hierarchy
- Einstein vs Einstein (Southampton)
Keynote | Echoes
- Extreme Feedback GEESE-ON
PDF | DM hierarchy
- IPM Apr 2025
PDF | Copenhagen Survey
- IPM Apr 2025
Keynote | Copenhagen Survey
- Kids Zoom Mar 2025
Keynote | Battle of the Big Bang
- Lee Fest 2025
Keynote | Cuscuton
- Matter
Keynote | General archive
- PSI 2025
Keynote | PSI
- QBH ASU 2025
Keynote | Quantum Black Holes
- QBH McMaster 2025
Keynote | Quantum Black Holes
- QBH Simons 2025
Keynote | Quantum Black Holes
- QBH Sydney 2025
Keynote | Quantum Black Holes
- QG Wild MannFest 2025
Keynote | Quadratic Gravity
- QG Wild MannFest 2025
PowerPoint | Quadratic Gravity
- QG Wild RRI 2025
Keynote | Quadratic Gravity
- Rahe Shiri
Keynote | public talk
- The Big Picture @UCL
Keynote | The Big Picture
- UV-Universe-QG-DIAS
PowerPoint | Quadratic Gravity
- UV-Universe-QG-DIAS
Keynote | Quadratic Gravity
- UV-Universe-QG-Imperial
Keynote | Quadratic Gravity
- UV-Universe-QG-Kings
Keynote | Quadratic Gravity
- What Banged
Keynote | public talk

2024 — 20 items

- 0202409
PDF | What I did during the last 17 months
- Chairing Seminars
Keynote | General archive
- concluding remarks Horndeski
Keynote | General archive
- ICBS Beijing 2024
Keynote | Precision Quantum Cosmology

- ICBS Beijing 2024
PDF | Precision Quantum Cosmology
- ICBS July 2024
Keynote | Precision Quantum Cosmology
- ICBS July 2024
PDF | Precision Quantum Cosmology
- Kyoto Feb 2024
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- Kyoto Feb 2024
Keynote | Precision Quantum Cosmology
- Niayesh Ramesh June 2024
Keynote | Abyss
- PSI 2024
Keynote | PSI
- QBH IHEP Beijing 2024
Keynote | Quantum Black Holes
- QBH UCAS Beijing 2024
Keynote | Quantum Black Holes
- QuRIOS Apr 2024
Keynote | Precision Quantum Cosmology
- RN50 talk template
Keynote | Abyss
- RN50 talk template
PowerPoint | Abyss
- Rome Feb 2024
Keynote | Endgame_colloquia
- Rome Feb 2024
PDF | Endgame_colloquia
- Tsinghua 2024
Keynote | Echo+Specs
- Tsinghua 2024
PDF | Echo+Specs

2023 — 26 items

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PDF | The Big Picture
- Afshordi N talk mac
Keynote | The Big Picture
- Krishan
Keynote | General archive
- Kruskal
Keynote | General archive
- Lord Byng Astro Club, May 2023
Keynote | public talk
- Niayesh WCA Jamboree 2023
Keynote | General archive
- Niayesh WCA Jamboree 2023
PDF | General archive
- Paradox Kitchenre Public Library 2023
Keynote | public talk
- Paradox Ottawa Zoom
Keynote | public talk
- Paradox Phys10 2023
Keynote | public talk
- PSI 2023
Keynote | PSI
- PSI 2023 1
Keynote | PSI
- QMUL interview 2023
Keynote | Job Interviews
- SFU Nov 2023
PDF | Echo+Specs
- SFU Nov 2023
Keynote | Endgame_colloquia
- SFU Nov 2023
PDF | Endgame_colloquia
- Sikkim Bani Dec 2023
Keynote | Endgame_colloquia
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- Strauss:Lupton Fest
Keynote | Strauss:Lupton Fest
- Testing Aspects of General Relativity-II Apr 2023
Keynote | Abyss
- Testing Gravity Vancouver Jan 2023
Keynote | Precision Quantum Cosmology
- The Big Picture @CASCA 2023
Keynote | The Big Picture
- The Big Picture @DEShaw
Keynote | The Big Picture
- UBC Jan 2023
Keynote | Endgame_colloquia
- Welcome
PDF | General archive
- Welcome
Keynote | General archive

2022 — 19 items

- BH to Big Bang
Keynote | BH_to_Big_Bang
- CNC Case anomaly
Keynote | CNC
- colloquia at UW
Keynote | General archive
- DM hierarchy KIAA Oct 13
Keynote | DM hierarchy
- DM hierarchy Mitchell May 14
Keynote | DM hierarchy
- DM hierarchy Sharif Oct 14
Keynote | DM hierarchy
- DM hierarchy UW Sep 14
Keynote | DM hierarchy
- Einstein vs Einstein (CERN)
Keynote | Echoes

- Einstein vs Einstein (IPM)
Keynote | Echoes
- Einstein vs Einstein (IPM)
PDF | Echoes
- Einstein vs Einstein (Testing GR 22)
Keynote | Echoes
- Fundamentalism
Keynote | Philosophy
- Fundamentalism
PDF | Philosophy
- PSI 2022

- Keynote | PSI
- QBH, Damghan
Keynote | Echoes
- Stimulating Hawking (GGI 2022)
Keynote | Echoes
- Stimulating Hawking (GGI 2022)
PDF | Echoes
- Stimulating Hawking (QSimFP 2022)
Keynote | Echoes
- Tips for Talks
Keynote | General archive

2021 — 23 items

- AltecsmoFun 2021
Keynote | Nothing
- Big Bang-Fact vs Fiction
Keynote | Big Bang
- Big Bang-Fact vs Fiction
PDF | Big Bang
- Black Holes, Unscripted
PDF | Echoes
- Black Holes, Unscripted
Keynote | Echoes
- Black Holes, Unscripted, Paris
Keynote | Echoes
- CnC MG 2021
Keynote | CNC
- CNC Strong Gravity
PDF | CNC
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Keynote | CNC
- CnC SUSY 2021
Keynote | CNC
- Cosmology Trials
Keynote | Big Bang
- Cosmology Trials PHYS 10

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- Covid IAP Mar 2021
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Keynote | Covid
- Covid UNI Apr 2021
Keynote | Covid
- e-Tek 2021
Keynote | public talk
- Einstein vs Einstein
PDF | Echoes
- Einstein vs Einstein Iran
Keynote | Echoes
- Einstein vs Einstein small
PDF | Echoes
- Gaia action streams21 2021
Keynote | Gaia
- Niayesh David 2021
Keynote | General archive
- Niayesh David 2021
PDF | General archive
- PSI 2021
Keynote | PSI

2020 — 47 items

- Abyss MSI McGill 2020
Keynote | Abyss
- BIU echoes
Keynote | Echoes
- BIU echoes
PDF | Echoes
- causal set conference 2020
PDF | General archive
- causal set conference 2020
Keynote | General archive
- CnC QG RG 2020
Keynote | CNC
- CnC QG RG 2020
PDF | CNC
- Covid Fields July 2020
Keynote | Covid

- Covid Fields July 2020
PDF | Covid
- Covid Google Sep 2020
Keynote | Covid
- Covid Iran SoA Sep 2020
Keynote | Covid
- Covid PI July 2020
Keynote | Covid
- Covid PI July 2020
PDF | Covid
- Covid York Oct 2020
Keynote | Covid
- D7
PDF | Covid
- Echo COVID 1
PDF | Echoes

- Echo COVID 2
PDF | Echoes
- Echoes status update PI Jan 2020
Keynote | Echoes
- For Brian 2020
Keynote | General archive
- Gaia action Canvas 2020
Keynote | Gaia
- Gaia action Penn 2020
Keynote | Gaia
- Gaia action Sharif 2020
Keynote | Gaia
- Gaia action Sharif 2020
PDF | Gaia
- Hubble Trouble QG Jun 2020
PDF | Gaia
- Hubble Trouble QG Jun 2020
Keynote | Nothing
- Hubble Trouble QG Jun 2020
PDF | Nothing
- I7
PDF | Covid
- Immunity
PDF | Covid
- Immunity
Keynote | Covid
- IPM echoes
Keynote | Echoes
- Jamboree 2020
PDF | General archive
- Jamboree 2020

Keynote | General archive

- Knowledge 2020
Keynote | The Nature of Scientific Knowledge
- Knowledge 2020
PDF | The Nature of Scientific Knowledge
- LISA echoes 2020
PDF | Echoes
- LISA echoes 2020
Keynote | Echoes
- NanoGravo+LIGO
Keynote | CNC
- Orleans
PDF | Covid
- PDT Echoes
Keynote | PDT
- PDT Echoes
PDF | PDT
- PstrainE2THlinearamplitudes v2
PDF | Echoes
- PstrainETHlinearamplitudes v2
PDF | Echoes
- Speceta50E2TH
PDF | Echoes
- Speceta50ETH
PDF | Echoes
- Untitled
Keynote | Covid
- Zanjan echoes
Keynote | Echoes
- Zanjan echoes
PDF | Echoes

2019 — 61 items

- AAA
PDF | Jamboree_2019
- Amber Hollinger
PDF | Jamboree_2019
- Andrej Obuljen
PDF | Jamboree_2019
- Andrew Reeves
PDF | Jamboree_2019
- Avery Broderick
PDF | Jamboree_2019
- Avery Broderick
Keynote | Jamboree_2019
- Bailey Robison
PDF | Jamboree_2019
- Boris Georgiev
PDF | Jamboree_2019
- Boris Georgiev
Keynote | Jamboree_2019
- Brian McNamara
PDF | Jamboree_2019
- Britt Jetter
PDF | Jamboree_2019

- Chengyu Xi
PDF | Jamboree_2019
- CnC BASIC 2019
PDF | CNC
- CnC BASIC 2019
Keynote | CNC
- Connor Martz
PDF | Jamboree_2019
- cosmo cruise 2019
PDF | Dark Energy
- cosmo cruise 2019
Keynote | Dark Energy
- Dustin Lang
PDF | Jamboree_2019
- Echoes TG19 Jan 2019
PDF | Echoes
- Echoes TG19 Jan 2019
Keynote | Echoes
- Faizan Mohammad
PDF | Jamboree_2019
- Gaia action PI 2019
Keynote | Gaia

- Ghazal Geshnizjani
PDF | Jamboree_2019
- GVSU colloquium
PDF | Nothing
- GVSU colloquium
Keynote | Nothing
- Hubble Trouble
Keynote | Nothing
- Isaac Spitzer
PDF | Jamboree_2019
- Jamboree 2019
PDF | Jamboree_2019
- Jamboree2019
Keynote | Jamboree_2019
- James Taylor
PDF | Jamboree_2019
- Karen McNab
PDF | Jamboree_2019
- Knowledge 2019
PDF | The Nature of Scientific Knowledge
- Knowledge 2019
Keynote | The Nature of Scientific Knowledge
- Krishan Saraswat
PDF | Jamboree_2019
- Kristi Webb
PDF | Jamboree_2019
- LIGO noise CnC
PDF | CNC
- Matthew Robbins
PDF | Jamboree_2019
- mcNab-jamboree-2018
PowerPoint | Jamboree_2019
- Michael Balogh
PDF | Jamboree_2019
- Michael Chapman
PDF | Jamboree_2019
- Mike Fich
PDF | Jamboree_2019
- Mike Hudson
PDF | Jamboree_2019
- Niayesh Afshordi
PDF | Jamboree_2019
- Nicole Drakos
PDF | Jamboree_2019
- OfDM
Keynote | DM hierarchy
- Picture McGill
Keynote | Power Spectra
- Prathamesh Tamhane
PDF | Jamboree_2019
- PSI 2019
Keynote | PSI
- Qingwen Wang
PDF | Jamboree_2019
- Quantum BH cartoon
Keynote | Echoes
- Quantum BH cartoon
PDF | Echoes
- Rahe Shiri
PDF | public talk
- Robb Mann
PDF | Jamboree_2019
- Robb Mann
PowerPoint | Jamboree_2019
- Supranta Bourah
PDF | Jamboree_2019
- Tianyi Yang
PDF | Jamboree_2019
- Tianyi Yang
Keynote | Jamboree_2019
- Will Percival
PDF | Jamboree_2019
- wrapup frontiers 2019
Keynote | General archive
- wrapup frontiers 2019
PDF | General archive
- ZZZ
PDF | Jamboree_2019

2018 — 62 items

- AA welcome
PDF | Jamboree_2018
- aether Philosophy summer inst jun 2018
Keynote | aether
- aether Philosophy summer inst jun 2018 2.29.44 AM
PDF | aether
- aether Western Mar 2018
Keynote | aether
- Aether WU March 2018
Keynote | aether
- Afshordi 2.29.45 AM
PDF | Jamboree_2018
- Altamirano
PDF | Jamboree_2018
- Babyk press
PDF | Jamboree_2018
- Balogh
PDF | Jamboree_2018
- Binder2
PDF | Jamboree_2018
- Binder3
PDF | Jamboree_2018
- Blind Spots Rotman Jun 2018
Keynote | DM hierarchy
- Blind Spots Rotman Jun 2018 2.29.44 AM
PDF | DM hierarchy
- Bobotsis - Astro Jamboree Slide
PDF | Jamboree_2018

- Broderick
PDF | Jamboree_2018
- CnC PI Asymptotic 2018
Keynote | CNC
- combined
PDF | Jamboree_2018
- ConnorMartz
PDF | Jamboree_2018
- Conrad Jamboree Winter2018
PDF | Jamboree_2018
- Dragons 2.29.43 AM
Keynote | public talk
- Dragons 2.29.44 AM
PDF | public talk
- Drakos AstroJamboree
PDF | Jamboree_2018
- Echoes McGill Apr 2018
Keynote | Echoes
- Echoes PI Feb 2018
Keynote | Echoes
- Echoes Solvay May 2018
Keynote | Echoes
- Echoes Western Mar 2018
Keynote | Echoes
- FichJamboree2018
PDF | Jamboree_2018
- Georgiev Astro Jamboree 2018
PDF | Jamboree_2018
- GW170817 BH
PDF | Echoes
- Hudson Jamboree
PDF | Jamboree_2018
- Jamboree2018
Keynote | Jamboree_2018
- Jamboree2018
PDF | Jamboree_2018
- Jamboree2018 1.24.46 AM
PDF | Jamboree_2018
- Jeter
PDF | Jamboree_2018
- Karami
PDF | Jamboree_2018
- Knowledge
Keynote | The Nature of Scientific Knowledge
- mann
PDF | Jamboree_2018
- McNamara
PDF | Jamboree_2018
- morawetz slide
PowerPoint | Jamboree_2018
- morawetz slide
PDF | Jamboree_2018
- Nanostructure CITA Jun 2018
Keynote | DM hierarchy
- Nanostructure CITA Jun 2018 2.29.44 AM
PDF | DM hierarchy
- Ni
PDF | Jamboree_2018
- Okoli
PDF | Jamboree_2018
- Percival 31-1-18
PDF | Jamboree_2018
- Preciado
PDF | Jamboree_2018
- PU
PDF | Jamboree_2018
- Reeves
PDF | Jamboree_2018
- Robbins - Astro Jamboree - January 31, 2018
PDF | Jamboree_2018
- Rufus
PowerPoint | Jamboree_2018
- SpitzerJamboree
PDF | Jamboree_2018
- Tamhane
PDF | Jamboree_2018
- Taylor
PDF | Jamboree_2018
- The Other 99 Percent
PDF | public talk
- The Other 99 Percent PI Jul 2018
Keynote | public talk
- Tiede a
PDF | Jamboree_2018
- Vantghem
PDF | Jamboree_2018
- Wang
PDF | Jamboree_2018
- webb jamboree slide
PDF | Jamboree_2018
- Xi
PDF | Jamboree_2018
- Yang
PDF | Jamboree_2018
- ZZ pizza
PDF | Jamboree_2018

2017 — 25 items

- Afshordi Echoes CosPA2017
PDF | Echoes
- Afshordi Echoes PAX 2017
PDF | Echoes
- CC many face
Keynote | CC_many_face
- Dark in extremes PI 2017
Keynote | DM hierarchy
- Dark in extremes Queens 2017
Keynote | DM hierarchy
- Echoes
PDF | Echoes

- Echoes CosPA2017
PDF | Echoes
- Echoes CosPA2017
Keynote | Echoes
- Echoes KU 2017
Keynote | Echoes
- Echoes OSU 2017
Keynote | Echoes
- Echoes OSU 2017
PDF | Echoes
- Echoes PAX 2017
Keynote | Echoes
- Echoes PI 2017
PDF | Echoes
- Echoes PI 2017
Keynote | Echoes
- Echoes UW 2017
Keynote | Echoes
- Echoes Vancouver 2017

- PDF | Echoes
- Echoes Vancouver 2017
Keynote | Echoes
- Fabric SISSA 2017
PDF | Reflections
- Fabric SISSA 2017
Keynote | Reflections
- Hologram
Keynote | public talk
- Hologram
PDF | public talk
- Refelction Western 2017
Keynote | Reflections
- Refelction Western 2017
PDF | Reflections
- summary Jing
Keynote | Echoes
- What Banged
PDF | public talk

2016 — 21 items

- 1108.0396v1
PDF | AiRE_disk
- 1309.5646v1
PDF | AiRE_disk
- AiRE disk DARK
PDF | AiRE_disk
- AiRE disk DARK
Keynote | AiRE_disk
- AiREGLQ (1)
PDF | AiRE_disk
- Alice PSI 2016
Keynote | Alice
- apj 779 2 95
PDF | AiRE_disk
- CC many face
PDF | CC_many_face
- CnC NBI
Keynote | CNC
- CnC NBI
PDF | CNC
- DM hierarchy McMaster Jun 16

- Keynote | DM hierarchy
- DM hierarchy McMaster Jun 16
PDF | DM hierarchy
- EFT CalTech 16
Keynote | EFT
- EFT CalTech 16 afshordi
PDF | EFT
- EFT CalTech 16 afshordi
Keynote | EFT
- neutrino florence
PDF | massive neutrinos
- neutrino florence
Keynote | massive neutrinos
- transitions1
PDF | AiRE_disk
- transitions2
PDF | AiRE_disk
- UW jamboree 16
Keynote | General archive
- UW jamboree 16
PDF | General archive

2015 — 21 items

- Afshordi CnC APS AnnArbor 2015
PDF | CNC
- Afshordi CNC Erlangen 2015
Keynote | CNC
- Afshordi CnC PI convergence 2015
Keynote | CNC
- Afshordi CNC SFU 2015
Keynote | CNC
- Afshordi CnC Theory Canada 2015
PDF | CNC

- Afshordi CnC Theory Canada 2015
Keynote | CNC
- Afshordi CNC Toronto 2015
Keynote | CNC
- Afshordi CNC Toronto 2015
PDF | CNC
- Alice Western 2015
Keynote | Alice
- Alice Western 2015
PDF | Alice

- BH to Big Bang
PDF | BH_to_Big_Bang
- CMB S4
Keynote | CNC
- CMB S4
PDF | CNC
- CnC Mannfest
Keynote | CNC
- CnC Mannfest
PDF | CNC
- Proof

Keynote | Proof!

- Proof
PDF | Proof!
- QG Plasma CnC
PDF | Cosmology_QG_Plasma
- QG Plasma CnC
Keynote | Cosmology_QG_Plasma
- Soap 2015
Keynote | public talk
- Welcome cosmic flows
Keynote | General archive

2014 — 12 items

- Afshordi CNC miami 2014
PDF | CNC
- Afshordi CNC miami 2014
Keynote | CNC
- CNC Case anomaly
PDF | CNC
- colloquia at UW
PDF | General archive
- DM hierarchy Mitchell May 14
PDF | DM hierarchy
- Horiozns 2014 Phys10
Keynote | public talk

- introduction universe sharif 2014
PowerPoint | public talk
- Jamboree UW 2014
PDF | General archive
- Jamboree UW 2014
Keynote | General archive
- Lorentz UTehran Oct 2014
Keynote | aether
- Perseids welcome 2014
Keynote | public talk
- Tips for Talks
PDF | General archive

2013 — 18 items

- BH shape dynamics
Keynote | aether
- BH shape dynamics
PDF | aether
- DM hierarchy PI Oct 13
Keynote | DM hierarchy
- DM hierarchy TAMU Mar 13
PowerPoint | DM hierarchy
- DM hierarchy TAMU Mar 13
PDF | DM hierarchy
- Eisntein Fail CASCA May 2013
Keynote | aether
- Eisntein Fail SuperJedi July 2013
Keynote | aether
- Jamboree PI 2013
Keynote | General archive
- Jamboree PI 2013
PDF | General archive

- Lorentz Madison Feb 2013
Keynote | aether
- Lorentz pic
PowerPoint | aether
- missing shanghai 2013
PDF | missing baryons
- missing shanghai 2013
Keynote | missing baryons
- SuperJedi galaxies 2013
PDF | General archive
- SuperJedi galaxies 2013
Keynote | General archive
- white hole miami dec 2013
Keynote | white hole
- white hole miami dec 2013
PDF | white hole
- white hole michigan sep 2013
Keynote | white hole

2012 — 16 items

- Aether PI May 2012
Keynote | aether
- Aether PI Oct 2012
Keynote | aether
- CDM Haloes

PDF | Cape Town Cosmo School 2012

- CDM Haloes
Keynote | Cape Town Cosmo School 2012
- DM hierarchy PI Sep 11
PowerPoint | DM hierarchy

- Einstein York Feb 2012
Keynote | aether
- Einstein York Feb 2012
PowerPoint | aether
- Geometry
PowerPoint | public talk
- Geometry EinsteinPlus July 2012
PowerPoint | public talk
- hot big bang
Keynote | HL_cosmology
- Jamboree Oct 2012

PDF | General archive

- Lifshitz
PDF | Cape Town Cosmo School 2012
- Lifshitz
Keynote | Cape Town Cosmo School 2012
- QG conference Oct 2012
Keynote | General archive
- UWC chair presentation
Keynote | UWC chair 2012
- UWC chair presentation
PDF | UWC chair 2012

2011 — 17 items

- aether APC June 2011
Keynote | aether
- CC philosophy UWO may 11
Keynote | aether
- Dark Cosmology
Keynote | Job Interviews
- DM ball
Keynote | General archive
- DM hierarchy Michigan Jan 11
PowerPoint | DM hierarchy
- DM hierarchy UNM May 11
PDF | DM hierarchy
- DM hierarchy UNM May 11
PowerPoint | DM hierarchy
- DM hierarchy UW Mar 11
PowerPoint | DM hierarchy
- Einstein CITA Sep 2011

Keynote | aether

- Einstein IUCAA Oct 2011
PDF | aether
- Einstein IUCAA Oct 2011
Keynote | aether
- Einstein KICP Oct 2011
Keynote | aether
- Gong Show Buffalo 2011
PowerPoint | General archive
- Gong Show Buffalo 2011
PDF | General archive
- gravity
PowerPoint | Gravity, the ultimate cosmological frontier
- LathamSACslides2011v6
PowerPoint | General archive
- rustbelt buffalo 2011 Jan 15
PowerPoint | aether

2010 — 17 items

- case low l 2010
PDF | ISW_anomaly
- case low l 2010
PowerPoint | ISW_anomaly
- CC philosophy PI may 10
PowerPoint | aether
- CITA25 May 2010
PowerPoint | aether
- DM hierarchy Case Feb 10
PowerPoint | DM hierarchy
- DM hierarchy IPM Aug 10
PowerPoint | DM hierarchy
- DM hierarchy Queens Mar 10
PowerPoint | DM hierarchy
- Gueph Colloquium Oct 19
PowerPoint | aether
- GWPI poster

PowerPoint | General archive

- GWPI poster
PDF | General archive
- IAP Feb 2010
PowerPoint | aether
- miami 2010 Dec 19
PowerPoint | aether
- ng
PowerPoint | General archive
- ng
PowerPoint | General archive
- Penn Feb 2010
PowerPoint | aether
- PSI 1
PowerPoint | public talk
- QG measurement ipm aug 10
PowerPoint | aether

2009 — 27 items

- **aether-Btt talk**
PDF | aether
- **Aspen Jan 09**
PowerPoint | aether
- **Case Aug 09**
PowerPoint | aether
- **dark cosmos**
PowerPoint | public talk
- **DM hierarchy KITPC Mar 09**
PowerPoint | DM hierarchy
- **DM hierarchy MIT Jun 09**
PDF | DM hierarchy
- **DM hierarchy MIT Jun 09**
PowerPoint | DM hierarchy
- **DM hierarchy MIT Jun 09**
PowerPoint | DM hierarchy
- **DM hierarchy Syracuse Mar 09**
PowerPoint | DM hierarchy
- **DM hierarchy UW Mar 09**
PowerPoint | DM hierarchy
- **IPMU Feb 09**
PowerPoint | aether
- **Iranian Caldendar**
PowerPoint | General archive
- **KITPC Mar 09**
PowerPoint | aether
- **miami 09**

- PowerPoint | aether
- **miami 09 afshordi**
PDF | aether
- **neutrinos and DE**
PowerPoint | Leiden_Apr_08
- **PI CC May 09**
PowerPoint | aether
- **PI job talk Mar 09**
PowerPoint | aether
- **PI theory canada Jun 09**
PowerPoint | aether
- **PSI 1**
PowerPoint | public talk
- **PSI 1**
PDF | public talk
- **PSI presentation**
PowerPoint | PSI
- **PSI presentation**
PDF | PSI
- **SFU Oct 09**
PowerPoint | aether
- **Stanford Sep 09**
PowerPoint | aether
- **UW jamboree**
PowerPoint | General archive
- **Vanderbilt Oct 09**
PowerPoint | aether

2008 — 21 items

- **aether**
PDF | aether
- **aether**
PowerPoint | aether
- **beyond horizon**
PowerPoint | Case-Western_08
- **beyond horizon Penn**
PowerPoint | Penn_job_tal_08
- **beyond horizon PI**
PowerPoint | PI_job_talk_08
- **beyond horizon Syracuse**
PowerPoint | syracuse
- **beyond horizon Syracuse**
PowerPoint | syracuse
- **DM hierarchy**
PowerPoint | DM hierarchy
- **Fundamental ISW**
PDF | Aspen_08
- **Fundamental ISW**
PowerPoint | Aspen_08
- **massive**

- PDF | massive gravity
- **massive**
PowerPoint | massive gravity
- **massive**
PowerPoint | massive gravity
- **mitseminar.pdf**
PDF | General archive
- **non-gaussianity**
PowerPoint | non-gaussianity
- **non-gaussianity sharif**
PowerPoint | non-gaussianity
- **SZ paradox**
PowerPoint | UW luster workshop July 08
- **szplane**
PowerPoint | syracuse
- **szplane toulouse**
PDF | toulouse
- **szplane toulouse**
PowerPoint | toulouse
- **what is wrong with cosmology**
PowerPoint | what is wrong with cosmology

2007 — 30 items

- **backreaction london**
PowerPoint | London_conference_07
- **caustics**
PowerPoint | Caustics
- **concent**
PowerPoint | concent
- **concent**
PDF | concent
- **cosmic baryonic budget**
PowerPoint | cluster reading group
- **cosmic baryonic budget**
PowerPoint | cluster reading group
- **Dark Side**
PowerPoint | Buffalo
- **Dark Side**
PDF | New Mexico
- **Dark Side**
PowerPoint | New Mexico
- **Dark Side**
PowerPoint | New Mexico
- **Dark Side**
PowerPoint | Pitt
- **Dark Side**
PowerPoint | UNC
- **Dark Side**
PowerPoint | York
- **digging**
PowerPoint | szmap
- **glast 07**
PowerPoint | Caustics

- **glast 07**
PowerPoint | Caustics
- **imprecision cosmology**
PowerPoint | Sharif_07
- **Journey Michigan**
PowerPoint | Michigan
- **Journey Michigan**
PowerPoint | Michigan
- **oxford interview**
PowerPoint | Oxford_interview
- **oxford talk**
PowerPoint | Oxford_interview
- **oxford talk**
PowerPoint | Oxford_interview
- **Research Plan**
PowerPoint | Pitt
- **szmap aspen**
PowerPoint | szmap
- **szmap london**
PDF | London_conference_07
- **szmap london**
PowerPoint | London_conference_07
- **szmap penn state**
PowerPoint | szmap
- **szplane**
PowerPoint | CITA
- **wmapII cosmology**
PowerPoint | princeton_apr07
- **wmapII cosmology**
PowerPoint | princeton_apr07

2006 — 8 items

- **caustics**
Web slides | Caustics
- **cuscuton**
PowerPoint | Cuscuton
- **dm windows**
PowerPoint | dm_windows
- **dm windows**
Web slides | dm_windows

- **szmap benasque**
PowerPoint | szmap
- **szmap fermilab**
PowerPoint | szmap
- **szmap irvine**
PowerPoint | szmap
- **szmap pi**
PowerPoint | szmap

2005 — 17 items

- **backreaction Chicago05 Poster**
PowerPoint | backreaction
- **backreaction MPA05**
PowerPoint | backreaction
- **backreaction MPA05 noanim**
PDF | backreaction
- **backreaction MPA05 noanim**
PowerPoint | backreaction
- **catast**
PowerPoint | catast
- **catast cosmo05**
PowerPoint | catast

- **catast cosmo05**
PDF | catast
- **cross talk columbia**
PDF | cross_talk
- **disk ins**
PowerPoint | disk_ins
- **Large Scale Structure**
PowerPoint | Qeshm_School_2005
- **Secondaries Observations**
PowerPoint | Qeshm_School_2005
- **Secondaries Theory**
PowerPoint | Qeshm_School_2005

- seminar at UFL
PowerPoint | backreaction
- shell
PDF | backreaction
- space

2004 — 15 items

- cross talk
Web slides | cross_talk
- cross talk AAS
PowerPoint | cross_talk
- cross talk aspen
PowerPoint | cross_talk
- cross talk columbia
PowerPoint | cross_talk
- cross talk isw
PowerPoint | cross_talk
- cross talk princeton
PowerPoint | cross_talk
- FPO
PowerPoint | FPO
- FPO

2003 — 11 items

- CROSS
PowerPoint | cross_poster
- cross talk
PowerPoint | cross_talk
- cross talk old
PowerPoint | cross_talk
- cross talk upenn
PowerPoint | cross_talk
- PBH
Web slides | Primordial Black Holes as Dark Matter
- PBH

2002 — 2 items

- Thin Accretion Disks
PowerPoint | Thin Accretion Disks

PDF | backreaction

- Thin Accretion Disks
PDF | Thin Accretion Disks
- thin disks
PDF | disk_ins

Web slides | FPO

- ISW
PowerPoint | Poster_Maryland
- slacconf two
PDF | SZ_map_Texas
- szmap
PDF | szmap
- szmap
PowerPoint | szmap
- Texas Poster
PowerPoint | SZ_map_Texas
- Texas Talk
PDF | SZ_map_Texas
- Texas Talk
PowerPoint | SZ_map_Texas

PDF | What I did during the last 12 months

- Primordial Black Holes as Dark Matter
PowerPoint | Primordial Black Holes as Dark Matter
- QPOs
Web slides | QPOs
- QPOs
PowerPoint | QPOs
- SZ
PowerPoint | SZ_sep_chicago
- What I did during the last 12 months
PowerPoint | What I did during the last 12 months

- What I did during the last 17 months
PowerPoint | What I did during the last 17 months